Linear Algebra Done Right

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1 Vector Space

1.1 Definition of Vector Space

Notation 1.1. $\mathbf{F}, V, W$
In this note, we use $\mathbf{F}$ to denote $\mathbf{R}$ or $\mathbf{C}$ and use V and W to denote vector spaces over $\mathbf{F}$.

Definition 1.2. Vector Space
A vector space is a set V over a field $\mathbf{F}$ along with two binary operators $+, \cdot$, satisfying the following properties.

1. **Closed** under addition and scalar multiplication
2. **Commutativity** of addition
3. **Associativity** of addition
4. **Associativity** of multiplication

$$(ab)v = a(bv) \text{ for all } a, b \in \mathbf{F} \text{ and all } v \in V$$

5. Existance of additive and multiplicative **identity**
6. Existance of additive **inverse**
7. **Distributivity** of vector sums

$$a(u + v) = au + av \text{ for all } a \in \mathbf{F} \text{ and all } u, v \in V$$

8. **Distributivity** of scalar sums

$$(a + b)v = av + bv \text{ for all } a, b \in \mathbf{F} \text{ and all } v \in V$$

Remark. Two binary operations are defined to be $+ : V \times V \to V$ and $\cdot : \mathbf{F} \times V \to V$, which contains the precondition of closure, i.e. the first property.

Notation 1.3. $\mathbf{F}^S$
If S is a set, then $\mathbf{F}^S$ denotes $\{f \mid f : S \to \mathbf{F}\}$

Remark. If S is a nonempty set, then $\mathbf{F}^S$ is a vector space over $\mathbf{F}$

1.2 Subspaces

Definition 1.4. Subspace
A subset U of V is called subspace of V if U is also a vector space under the same addition and multiplication.

Remark. A subset U of V is a subspace iff U satisfies the following conditions:

1. Existance of additive **identity**
2. **Closed** under addition and scalar multiplication

Definition 1.5. Sum of Subsets
Suppose $U_1, \cdots, U_m$ are subsets of V. Then the sum of $U_1, \ldots, U_m$, denoted $U_1 + \cdots + U_m$, is defined as

$$U_1 + \cdots + U_m = \{u_1 + \cdots + u_m : u_1 \in U_1, \ldots, u_m \in U_m\}$$

Remark. Kind of like **convolution**.

Proposition 1.6. Suppose $U_1, \ldots, U_m$ are subspaces of V. Then $U_1 + \cdots + U_m$ is the smallest subspaces of V containing $U_1, \ldots, U_m$ (meaning other subspaces containing $U_1, \ldots, U_m$ also contains $U_1 + \cdots + U_m$).

1.3 Direct Sums

Definition 1.7. Direct Sum
Suppose $U_1, \ldots, U_m$ are subspaces of V. Then the sum $U_1 + \cdots + U_m$ is called a direct sum iff any element in $U_1 + \cdots + U_m$ can be written in only one way as a sum $u_1 + \cdots + u_m$, where $u_i \in U_i$.

Notation 1.8. $U_1 \oplus U_2$
We use $U_1 \oplus \cdots \oplus U_m$ to denote that $U_1 + \cdots + U_m$ is a direct sum.

Proposition 1.9. Suppose $U_1, \ldots, U_m$ are subspaces of V. Then $U_1 + \cdots + U_m$ is a direct sum iff there's only one way to write 0 as a sum $u_1 + \cdots + u_m$.

Remark. The only way is by taking each of $u_1, \ldots, u_m$ equal to 0.

Proposition 1.10. Suppose U and W are subspaces of V. Then $U + W$ is a direct sum iff $U \cap W = \{0\}$.

2 Finite-Dimensional Vector Spaces

2.1 Span and Linear Independence

Definition 2.1. Span

$$\text{span}(v_1, \ldots, v_m) = \{a_1 v_1 + \cdots + a_m v_m \mid a_1, \ldots, a_m \in \mathbf{F}\}$$
$$\text{span}() = \{0\}$$

Remark. Span is the collection of linear combinations.

Proposition 2.2. The span of a list of vectors in V is the smallest subspace of V containing all the vectors in the list.

Remark. Compared with Proposition 1.6.

Definition 2.3. Spans
If $\text{span}(v_1, \ldots, v_m)$ equals V, we say that $v_1, \ldots, v_m$ spans V.

Definition 2.4. Finite & Infinite-dimensional vector space
A vector space is called finite-dimensional iff some list of vectors in it spans the space.
A vector space is called infinite-dimensional if it is not finite-dimensional.

Notation 2.5. $\mathcal{P}(\mathbf{F})$
We use $\mathcal{P}(\mathbf{F})$ to represent the set of all polynomials with coefficients in $\mathbf{F}$, i.e.

$$\mathcal{P}(\mathbf{F}) = \left\{p : \mathbf{F} \to \mathbf{F} \mid p(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_m z^m, a_0, \ldots, a_m \in \mathbf{F}\right\}$$

Remark. We can regard $p(z)$ as a list of $m + 1$ vectors.

Remark. Note that $\mathcal{P}(\mathbf{F})$ is a subspace of $\mathbf{F}^{\mathbf{F}}$, as defined in Notation 1.3.

Remark. Also note that $\mathcal{P}(\mathbf{F})$ is infinite-dimensional.

Notation 2.6. $\deg p$
For a polynomial $p \in \mathcal{P}(\mathbf{F})$, we say $\deg p = m$ if

$$p(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_m z^m$$

for all $z \in \mathbf{F}$ and $a_m \neq 0$.
For $p(z) = 0$, we define $\deg p = -\infty$.

Notation 2.7. $\mathcal{P}_m(\mathbf{F})$
For $m \in \mathbf{N}$, $\mathcal{P}_m(\mathbf{F})$ denotes the set of all polynomials with coefficients in $\mathbf{F}$ and degree at most m.

Remark. Note that $\mathcal{P}_m(\mathbf{F})$ is finite-dimensional.

Definition 2.8. Linearly Independent & Dependent
A list $v_1, \ldots, v_m$ of vectors in V is called linearly independent iff

$$a_1 v_1 + \cdots + a_m v_m = 0 \iff a_1 = a_2 = \cdots = a_m = 0$$

We also define the empty list () to be linearly independent.

A list is linearly dependent iff it's not linearly independent.

Remark. If $v_1, \ldots, v_m$ is linearly independent, then each vector in $\operatorname{span}(v_1, \ldots, v_m)$ has only one representation as a linear combination of $v_1, \ldots, v_m$.

Example 2.9. Special Case

- A list v is linearly independent iff $v \neq 0$.
- A list v_1, v_2 is linearly independent iff $\nexists a \in \mathbf{F}, v_1 = a v_2$.
- The list $1, z, \ldots, z^m$ is linearly independent in $\mathcal{P}(\mathbf{F})$ for all $m \in \mathbf{N}$.
- Removing any element in a list doesn't change its linearly independency.

- If some vector in a list of vectors in V is a linear combination of the other vectors, then the list is linearly dependent.
- Every list containing the 0 vector is linearly dependent.

Lemma 2.10. Suppose $v_1, \ldots, v_m$ is a linearly dependent list in V, then there exists $j \in [1, m]$ such that:

1. $v_j \in \text{span}(v_1, \ldots, v_{j-1})$
2. If the jth term is removed from $v_1, \ldots, v_m$, the span of the remaining list equals $\text{span}(v_1, \ldots, v_m)$

Remark. This lemma is really useful in constructing and judging linearly independent list.

Theorem 2.11. In a finite-dimensional vector space V, if a list of vectors $v_1, \ldots, v_m$ are linearly independent, and another list of vectors $u_1, \ldots, u_n$ spans V, then $m \leq n$.

Remark. The smallest list which spans a finite-dimensional vector space is linearly independent. And that list is the biggest list that is linearly independent.

Theorem 2.12. Every subspace of a finite-dimensional vector space is finite-dimensional.

Question. Suppose $v_1, \ldots, v_m$ is linearly independent in V, and each of them can be uniquely represented as linear combinations of $u_1, \ldots, u_n$. Then, $u_1, \ldots, u_n$ is linearly independent?

2.2 Bases

Definition 2.13. Basis
A basis of V is a list of vectors in V which is linearly independent and spans V.

Proposition 2.14. A list $v_1, \ldots, v_n$ is a basis of V iff every $v \in V$ can be written uniquely in the form

$$v = a_1 v_1 + \cdots + a_n v_n$$

where $a_1, \ldots, a_n \in \mathbf{F}$.

Proposition 2.15. Every list which spans V contains a basis of V.

Remark. Consider Lemma 2.10.

Proposition 2.16. Every linearly independent list in a finite-dimensional vector space is contained in a basis of V.

Remark. If a linearly independent list is added a vector which is not in its previous span, then after addition, the list remains linearly independent. Then by Theorem 2.11 there's an upper bound and we can extend it this way.

Proposition 2.17. Every finite-dimensional vector space has a basis.

Remark. Referring to Definition 2.4 and Proposition 2.15.

Proposition 2.18. Suppose V is finite-dimensional and U is a subspace of V. Then there is a subspace W of V such that $V = U \oplus W$.

2.3 Dimension

Definition 2.19. Dimension
The dimension of a finite-dimensional vector space is the length of any basis of the vector space. The dimension of $\{0\}$ is defined to be 0.

Notation 2.20. $\dim V$
We use $\dim V$ to denote the dimension of V.

Remark. It can be prove that any two basis of the same vector space have the same length.

Example 2.21.

- $\dim \mathbf{F}^n = n$
- $\mathcal{P}_m(\mathbf{F}) = m + 1$

Proposition 2.22. If V is finite-dimensional and U is the subspace of V, then $\dim U \leq \dim V$.

Remark. In this proposition, $\dim U = \dim V \iff U = V$.

Proposition 2.23. If V is finite-dimensional, then every linearly independent list of vectors in V with length $\dim V$ is a basis of V.

Remark. We can try to extend the list by Proposition 2.16 but the the list has the same length of the basis. It is similar to the proposition that

$$A \subseteq B \ \& \ |A| = |B| \in \mathbf{N} \implies A = B$$

Proposition 2.24. If V is finite-dimensional, then every spanning list of vectors in V with length $\dim V$ is a basis of V.

Proposition 2.25. If U_1 and U_2 are subspaces of a finite-dimensional vector space, then

$$\dim(U_1 + U_2) = \dim U_1 + \dim U_2 - \dim(U_1 \cap U_2)$$

Remark. Similar to the proposition that

$$|U_1 \cup U_2| = |U_1| + |U_2| - |U_1 \cap U_2|$$

and we replace $U_1 \cup U_2$ with $U_1 + U_2$ to make it a vector space.

3 Linear Maps

3.1 The Vector Space of Linear Maps

Definition 3.1. Linear map
A linear map from V to W is a function $T : V \to W$ with the following properties:

- **Additivity**
$$T(u + v) = T(u) + T(v) \text{ for all } u, v \in V$$

- **Homogeneity**

$$T(\lambda v) = \lambda T(v) \text{ for all } \lambda \in \mathbf{F} \text{ and all } v \in V$$

Notation 3.2. $\mathcal{L}(V, W)$
The set of all linear maps from V to W is denoted $\mathcal{L}(V, W)$.

Example 3.3.

- **zero**: Define $0 \in \mathcal{L}(V, W)$ by

$$0(v) = 0$$

- **identity**: Define $I \in \mathcal{L}(V, W)$ by

$$I(v) = v$$

- **differentiation**: Define $D \in \mathcal{L}(\mathcal{P}(\mathbf{R}), \mathcal{P}(\mathbf{R}))$ by

$$D(p) = p'$$

- **integration**: Define $T \in \mathcal{L}(\mathcal{P}(\mathbf{R}), \mathbf{R})$ by

$$T(p) = \int_0^1 p(x)\mathrm{d}x$$

- **backward shift**: Define $T \in \mathcal{L}(\mathbf{F}^\infty, \mathbf{F}^\infty)$ by

$$T(x_1, x_2, x_3, \dots) = (x_2, x_3, \dots)$$

- **from $\mathbf{F}^n$ to $\mathbf{F}^m$**: every $T \in \mathcal{L}(\mathbf{F}^n, \mathbf{F}^m)$ is defined by

$$T(x_1, \dots, x_n) = (\sum_{i=1}^{n} A_{1,i}x_i, \dots, \sum_{i=1}^{n} A_{m,i}x_i)$$

Proposition 3.4. Suppose $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_n$ is a basis of W. Then there exists a unique linear map $T : V \to W$ such that

$$T(v_i) = w_i \text{ for all } i \in \{1, \dots, n\}$$

Remark. Every value in V can be represented by linear combination of basis of V, and by the properties of T, the function value is uniquely determined.

Proposition 3.5. $\mathcal{L}(V, W)$ is a vector space with addition and scalar multiplication defined by

$$(S+T)(v) = S(v) + T(v)$$
$$(\lambda T)(v) = \lambda T(v)$$

for all $S, T \in \mathcal{L}(V, W)$, $\lambda \in \mathbf{F}$ and $v \in V$

Definition 3.6. Product of Linear Maps
For $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, then their product $ST \in \mathcal{L}(U, W)$ is defined by

$$(ST)(u) = S(T(u))$$

for all $u \in U$

Remark. The same definition as their composition $S \circ T$.

Proposition 3.7. The product of linear maps has the following properties:

- **associativity**:
$$(T_1T_2)T_3 = T_1(T_2T_3)$$
- **identity**:
$$TI = IT = T$$
- **distributivity**:
$$(S_1 + S_2)T = S_1T + S_2T$$
$$S(T_1 + T_2) = ST_1 + ST_2$$

Proposition 3.8. Suppose T is a linear map, then $T(0) = 0$.

Remark. Consider $T(0) = T(0) + T(0)$

Definition 3.9. Null Space
For $T \in \mathcal{L}(V, W)$, the null space of T is defined to be $\{v \in V \mid T(v) = 0\}$

Notation 3.10. null T
We use null T to denote the null space of T.

Remark. Note that 0 is in the null space of every linear map (by Proposition 3.8).

Proposition 3.11. Suppose $T \in \mathcal{L}(V, W)$, then null T is a subspace of V.

Definition 3.12. Injective
A function $T : V \to W$ is called injective iff $T(u) = T(v) \implies u = v$

Remark. A necessary and sufficient condition for injectivity would be null $T = \{0\}$

Definition 3.13. Range
For $T \in \mathcal{L}(V, W)$, the range of T is defined to be $\{T(v) \mid v \in V\}$

Notation 3.14. range T
We use range T to denote the range of T.

Remark. $0 \in$ range T by Proposition 3.8 .

Remark. Note that for $T \in \mathcal{L}(V, W)$, range T is a subspace of W

Definition 3.15. Surjective
A function $T \in \mathcal{L}(V, W)$ is called surjective iff range $T = W$

Remark. T in injective iff range $T = V$.

Theorem 3.16. Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Then range T is finite-dimensional and we have

$$\dim V = \dim \text{null } T + \dim \text{range } T$$

Remark. This is the **Fundamental Theorem of Linear Maps**.

Corollary 3.17. If $\dim V > \dim U$, then no linear map from V to W is injective.

Corollary 3.18. If $\dim V < \dim U$, then no linear map from V to W is surjective.

Proposition 3.19. A homogeneous system of linear equations with more variables than equations has nonzero solutions.

Remark. A homogeneous system of linear equations is in the form of

$$\begin{cases} \sum_{i=1}^{n} A_{1,i}x_i = 0 \\ \qquad \vdots \\ \sum_{i=1}^{n} A_{m,i}x_i = 0 \end{cases}$$

Then consider an equivalent function $T : \mathbf{F}^n \to \mathbf{F}^m$ defined by

$$T(x_1, \ldots, x_n) = (\sum_{i=1}^{n} A_{1,i}x_i, \ldots, \sum_{i=1}^{n} A_{m,i}x_i)$$

Therefore, $n > m$ implies $\dim \mathbf{F}^n > \dim \mathbf{F}^m$, and thus T is not injective by Corollary 3.17, which means there are nonzero solutions.

Proposition 3.20. An inhomogeneous system of linear equations with more equations than variables has no solution for some choice of the constant terms.

Remark. An inhomogeneous system of linear equations is in the form of

$$\begin{cases} \sum_{i=1}^{n} A_{1,i}x_i = c_1 \\ \qquad \vdots \\ \sum_{i=1}^{n} A_{m,i}x_i = c_m \end{cases}$$

Then rephrase the system by the same function in Proposition 3.19. Hence, $n < m$ implies T is not surjective.

3.2 Matrices

Definition 3.21. Matrix
An m-by-n matrix A is a rectangular array of elements of $\mathbf{F}$ with m rows and n columns, where $A_{i,j}$ denotes the entry in row i and column j of A.

$$A = \begin{pmatrix} A_{1,1} & \dots & A_{1,n} \\ \vdots & & \vdots \\ A_{m,1} & \dots & A_{m,n} \end{pmatrix}$$

Definition 3.22. Matrix of a linear map
Suppose $T \in \mathcal{L}(V, W)$ and $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_m$ is a basis of W. Then the matrix of T with respect to these bases is the m-by-n matrix $\mathcal{M}(T)$ whose entries $A_{i,j}$ are defined by

$$T(v_j) = A_{1,j}w_1 + \cdots + A_{m,j}w_j$$

That means, we have

$$\mathcal{M}(T) = \begin{array}{c} \\ w_1 \\ \vdots \\ w_m \end{array} \begin{array}{c} \begin{array}{cccc} v_1 & \dots & v_k & \dots & v_n \end{array} \\ \begin{pmatrix} & & A_{1,k} & & \\ & & \vdots & & \\ & & A_{m,k} & & \end{pmatrix} \end{array}$$

We also use the notation $\mathcal{M}(T, (v_1, \dots, v_n), (w_1, \dots, w_m))$ to clearify the bases.

Remark. In fact, matrix is similar to linear map to some extent, and we call this relationship isomorphic in Proposition 3.38.

Definition 3.23. Matrix Addition
For matrix A, B with the same size, $A + B$ is defined by

$$(A + B)_{i,j} = A_{i,j} + B_{i,j}$$

Proposition 3.24. Suppose $S, T \in \mathcal{L}(V, W)$, then $\mathcal{M}(S + T) = \mathcal{M}(S) + \mathcal{M}(T)$.

Definition 3.25. Scalar multiplication of a matrix
For matrix A and $\lambda \in \mathbf{F}$, λA is defined by

$$(\lambda A)_{i,j} = \lambda A_{i,j}$$

Proposition 3.26. Suppose $\lambda \in \mathbf{F}$ and $T \in \mathcal{L}(V, W)$, then $\mathcal{M}(\lambda T) = \lambda \mathcal{M}(T)$.

Notation 3.27. $\mathbf{F}^{m,n}$
The set of all m-by-n matrices with entries in $\mathbf{F}$ is denoted by $\mathbf{F}^{m,n}$.

Remark. Recall that for $n \in \mathbf{Z}^+$, $\mathbf{F}^n$ denotes all **lists** of length n of elements in F, and for set S, $\mathbf{F}^S$ denotes all **functions** from S to $\mathbf{F}$.

Proposition 3.28. With addition and scalar multiplication defined above, $\mathbf{F}^{m,n}$ is a vector space with dimension mn.

Definition 3.29. Matrix multiplication
Suppose A is an m-by-n matrix and C is an n-by-p matrix. Then AC is defined to be an m-by-p matrix such that

$$(AC)_{i,j} = \sum_{r=1}^{n} A_{i,r} C_{r,j}$$

Remark. This definition can ensure that $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$.

Notation 3.30. $A_{j,\cdot}$ & $A_{\cdot,k}$
For an m-by-n matrix A, $A_{j,\cdot}$ denotes the 1-by-n matrix consisting of row j of A, and $A_{\cdot,k}$ denotes the m-by-1 matrix consisting of colomn k of A

Remark. Based on this notation, we have $(AC)_{j,k} = A_{j,\cdot} C_{\cdot,k}$

Proposition 3.31. Suppose A is an m-by-n matrix and C is an n-by-p matrix. Then

$$(AC)_{\cdot,k} = AC_{\cdot,k}$$

3.3 Invertibility and Isomorphic Vector Spaces

Definition 3.32. Invertible & Inverse
A linear map $T \in \mathcal{L}(V, W)$ is called invertible iff there exists a linear map $S \in \mathcal{L}(W, V)$ such that

$$ST = TS = I$$

Then S is called an inverse of T.

Proposition 3.33. An invertible linear map has a unique inverse.

Notation 3.34. T^{-1}
Suppose T is invertible, then we use T^{-1} to denote the inverse of T.

Remark. T^{-1} is well defined since the inverse is unique.

Proposition 3.35. A linear map is invertible iff it is injective and surjective.

Remark. For $f(v) = w$, injectivity means the **uniqueness** of v and surjectivity means the **existence** w.

Definition 3.36. Isomorphism & Isomorphic
An isomorphism is an invertible linear map.
Two vector spaces are called isomorphic if there is an isomorphism from one vector space onto the other one.

Proposition 3.37. Two finite-dimensional vector spaces over $\mathbf{F}$ are isomorphic iff they have the same dimension.

Remark. For $T \in \mathcal{L}(V, W)$,
If T is injective, then $\dim V \leq \dim U$
If T is surjective, then $\dim V \geq \dim U$
If T is invertible, then $\dim V = \dim U$

Proposition 3.38. Suppose $\dim V = n$ and $\dim W = m$, then $\mathcal{L}(V, W)$ and $\mathbf{F}^{m,n}$ are isomorphic.

Remark. $\mathcal{M}(T)$ is an isomorphism between them.

Proposition 3.39. Suppose V, W are finite-dimensional, then $\dim \mathcal{L}(V, W)$ is also finite-dimensional and

$$\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$$

Remark. This can be proved by Proposition 3.28, Proposition 3.37 and Proposition 3.38.

Definition 3.40. Matrix of a vector
Suppose $v \in V$ and $v_1, \ldots, v_n$ is a basis of V. Then the matrix of v with respect to this basis is the n-by-1 matrix defined by

$$\mathcal{M}(v) = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}$$

where

$$v = c_1 v_1 + \cdots + c_n v_n$$

Remark. Here $\mathcal{M}$ is also an isomorphism from V to $\mathbf{F}^{n,1}$, building the connection between vector and matrix.

Proposition 3.41. For $T \in \mathcal{L}(V, W)$ and $v_1, \ldots, v_n$ is a basis of V, then

$$\mathcal{M}(T)_{\cdot,k} = M(v_k)$$

Proposition 3.42. For $T \in \mathcal{L}(V, W)$ and $v \in V$, we have

$$\mathcal{M}(T(v)) = \mathcal{M}(T)\mathcal{M}(v)$$

Remark. Here we make connections between linear map and matrix multiplication.

Definition 3.43. Operator
A linear map from a vector space V to itself is called an operator, denoted $\mathcal{L}(V) = \mathcal{L}(V, V)$.

Theorem 3.44. Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then the following are equivalent:

- T is injective.
- T is surjective.
- T is invertible.

Remark. Consider Theorem 3.16.

Remark. The equivalence doesn't hold for infinite-dimensional V.

Remark. This reveals a "symmetry" of finite-dimensional vector space.

Example 3.45. For each polynomial $q \in \mathcal{P}(\mathbf{R})$, there exists a polynomial $p \in \mathcal{P}(\mathbf{R})$ with $((2x^2 + 3x + 7)p)'' = q$.

Proof. For each $q \in \mathcal{P}(\mathbf{R})$, suppose $\deg q = m$, then define $T : \mathcal{P}_m(\mathbf{R}) \to \mathcal{P}_m(\mathbf{R})$ by

$$T(p) = ((2x^2 + 3x + 7)p)''$$

Obviously null $T = 0$, thus T is injective, and then, by Theorem 3.44, T is surjective, which implies the existance of p such that $T(p) = q$. □

3.4 Products and Quotients of Vector Spaces

Definition 3.46. Product of vector spaces
Suppose $V_1, \ldots, V_m$ are vector spaces over $\mathbf{F}$, then their product is a vector space defined by

$$V_1 \times \cdots \times V_m = \{(v_1, \ldots, v_m) \mid v_i \in V_i\}$$

with addition and multiplication on the space defined by

$$(u_1, \ldots, u_m) + (v_1, \ldots, v_m) = (u_1 + v_1, \ldots, u_m + v_m)$$

$$\lambda(v_1, \ldots, v_m) = (\lambda v_1, \ldots, \lambda v_m)$$

Example 3.47. $\mathbf{R}^2 \times \mathbf{R}^3$ isn't equal to $\mathbf{R}^5$, but isomorphic to $\mathbf{R}^5$.

Proposition 3.48. Suppose $V_1, \ldots, V_m$ are finite-dimensional vector spaces, then $V_1 \times \cdots \times V_m$ is finite-dimensional and we have

$$\dim(V_1 \times \cdots \times V_m) = \dim V_1 + \cdots + \dim V_m$$

Remark. Suppose a basis of V_i is $v_{i,1}, \ldots, v_{i,s_i}$ for all $i \in \{1, \ldots, m\}$, then a basis of $V_1 \times \cdots \times V_m$ is $\{(0, \ldots, 0, v_{i,j}, 0, \ldots, 0) \mid 1 \leq i \leq m, 1 \leq j \leq s_i\}$

Proposition 3.49. Suppose $U_1, \ldots, U_m$ are subspaces of V. Define a linear map $\Gamma : U_1 \times \cdots \times U_m \to U_1 + \cdots + U_m$ by

$$\Gamma(u_1, \ldots, u_m) = u_1 + \cdots + u_m$$

Then Γ is surjective. And Γ is injective iff $U_1 + \cdots + U_m$ is a direct sum.

Remark. This reveals the product and direct sum of vector spaces are isomorphic.

Corollary 3.50. Suppose V is finite-dimensional and $U_1 \ldots, U_m$ are subspaces of V. Then $U_1 + \cdots + U_m$ is a direct sum iff

$$\dim(U_1 + \cdots + U_m) = \dim U_1 + \cdots + \dim U_m$$

Definition 3.51. $v + U$
Suppose $v \in V$ and U is a subspace of V, then $v + U$ is a subset of V defined by

$$v + U = \{v + u \mid u \in U\}$$

Definition 3.52. Affine subset
An affine subset of V is a subset of V of the form $v + U$ for some $v \in V$ and some subspace U of V.

Definition 3.53. Parallel
For $v \in V$ and U a subset of V, the affine subset $v + U$ is said to be parallel to U.

Definition 3.54. Quotient space
Suppose U is a subspace of V, then the quotient space V/U is the set of all affine subsets of V parallel to U, which means

$$V/U = \{v + U \mid v \in V\}$$

Proposition 3.55. Suppose U is a subspace of V and $v, w \in V$, then the following are equivalent:

- $v - w \in U$
- $v + U = w + U$
- $(v + U) \cap (w + U) \neq \varnothing$

Proposition 3.56. Suppose U is a subspace of V, then V/U is a vector space, with the operations of addition and scalar multiplication defined by

$$(v + U) + (w + U) = (v + w) + U$$

$$\lambda(v + U) = (\lambda v) + U$$

Remark. Under such definition, $v \in V$ and $v + U \in V/U$ are correspondingly related in form. But note that $v + U$ can repeat but v cannot, thus two spaces are not isomorphic.

Definition 3.57. Quotient map
Suppose U is a subspace of V, then the quotient map π is linear map $\pi : V \to V/U$ defined by

$$\pi(v) = v + U$$

Proposition 3.58. Suppose V is a finite-dimensional and U is a subspace of V. Then

$$\dim V/U = \dim V - \dim U$$

Remark. Proved by null $\pi = U$.

Definition 3.59. $\tilde{T}$
Suppose $T \in \mathcal{L}(V, W)$. Define $\tilde{T} : V/(\text{null } T) \to W$ by

$$\tilde{T}(v + \text{null } T) = T(v)$$

Remark. $\tilde{T}$ is a "compression" of T which puts together w and v satisfying $T(w) = T(v)$. Actually, we have

$$T(u) = \tilde{T}(U) \iff u \in U.$$

Proposition 3.60. Suppose $T \in \mathcal{L}(V, W)$, then

- $\tilde{T}$ is a linear map from $V/(\text{null } T)$ to W.
- $\tilde{T}$ is injective.
- range $\tilde{T}$ = range T.
- $V/(\text{null } T)$ is isomorphic to range T.

3.5 Duality

Definition 3.61. Linear functional
A linear functional on V is a linear map from V to $\mathbf{F}$.

Remark. By Theorem 3.16, for a nonzero linear functional $T : V \to \mathbf{F}$, we have $\dim \text{null } T = \dim V - 1$.

Definition 3.62. Dual space
The dual space of V, denoted V', is the vector space of all linear functionals on V, i.e. $V' = \mathcal{L}(V, \mathbf{F})$.

Proposition 3.63. Suppose V is finite-dimensional, then V' is also finite-dimensional and $\dim V' = \dim V$.

Remark. Special case of Proposition 3.39.

Definition 3.64. Dual basis
If $v_1, \ldots, v_n$ is a basis of V, then the dual basis of $v_1, \ldots, v_n$ is the list $\varphi_1, \ldots, \varphi_n$ of elements of V', where each φ_j is the linear functional on V such that

$$\varphi_j(v_k) = \begin{cases} 1 & k = j \\ 0 & k \neq j \end{cases}$$

Remark. φ_j represents the linear function that **selects** the jth coordinate, and $\text{span}(\varphi_j, \varphi_k)$ represents the set of linear combination of the jth and k^{th} coordinate.

Proposition 3.65. Suppose V is finite-dimensional. Then the dual basis of a basis of V is a basis of V'.

Definition 3.66. Dual map
If $T \in \mathcal{L}(V, W)$, then the dual map of T is the linear map $T' \in \mathcal{L}(W', V')$ defined by $T'(\varphi) = \varphi \circ T$ for $\varphi \in W'$.

Remark. $\varphi \in W'$ is the linear combination of the basis of W, and $T \in \mathcal{L}(V, W)$ makes every component of basis of W to be represented as linear combinations of the basis of V, thus T' seems to change the domain of φ from W to V.

Remark.

- For a vector space V, V' denotes the dual space of V.
- For a linear map T, T' denotes the dual map of T.
- For a polynomial p, p' denotes the derivative of p.

Proposition 3.67.

- $(S + T)' = S' + T'$ for all $S, T \in \mathcal{L}(V, W)$.
- $(\lambda T)' = \lambda T'$ for all $\lambda \in \mathbf{F}$ and all $T \in \mathcal{L}(V, W)$.
- $(ST)' = T'S'$ for all $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$.

Remark. The first two results indicate that the function which takes T to T' is a linear map.

Definition 3.68. Annihilator
For $U \subset V$, the annihilator of U, denoted U^0, is defined by

$$U^0 = \{\varphi \in V' \mid \varphi(u) = 0,\ \forall u \in U\}$$

Example 3.69. Let e_1, e_2, e_3, e_4, e_5 denote the standard basis of $\mathbf{R}^5$, and let $\varphi_1, \varphi_2, \varphi_3, \varphi_4, \varphi_5$ denote the dual basis of $(\mathbf{R}_5)'$. Suppose $U = \text{span}(e_1, e_2)$, then $U^0 = \text{span}(\varphi_3, \varphi_4, \varphi_5)$.

Proposition 3.70. Suppose $U \subset V$. Then U^0 is a subspace of V'.

Proposition 3.71. Suppose V is finite-dimensional and U is a subspace of V. Then

$$\dim U + \dim U^0 = \dim V$$

Remark. Consider $i \in \mathcal{L}(U, V)$ defined by $i(u) = u$, then $i' \in \mathcal{L}(V', U')$, thus by Theorem 3.16

$$\dim \text{range } i' + \dim \text{null } i' = \dim V$$

then we prove it by null $i' = U^0$ and range $i' = U'$.

$$i'(\varphi) = 0 \iff \forall u \in U, i'(\varphi)(u) = 0 \iff \forall u \in U, \varphi(i(u)) = 0 \iff \varphi \in U^0$$

Proposition 3.72. Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$, then

- null $T' = (\text{range } T)^0$
- $\dim \text{null } T' = \dim \text{null } T + \dim W - \dim V$

Proposition 3.73. Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then T is surjective iff T' is injective.

Proposition 3.74. Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then

- $\dim \text{range } T' = \dim \text{range } T$
- range $T' = (\text{null } T)^0$

Proposition 3.75. Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then T is injective iff T' is surjective.

Definition 3.76. Transpose
Suppose A is an m-by-n matrix, then A^{t} is the n-by-m matrix whose entries are given by

$$A^{\mathrm{t}}_{i,j} = A_{j,i}$$

Proposition 3.77. If A is an m-by-n matrix and C is an n-by-p matrix, then

$$(AC)^{\mathrm{t}} = C^{\mathrm{t}} A^{\mathrm{t}}$$

Proposition 3.78. Suppose $T \in \mathcal{L}(V, W)$. Then $\mathcal{M}(T') = (\mathcal{M}(T))^{\mathrm{t}}$

Remark. As mentioned above, T' is like representing the basis of W as linear combination of the basis of V. The transpose has similar meaning as well.

Definition 3.79. Row Rank & Column Rank
Suppose A is an m-by-n matrix with entries in $\mathbf{F}$. Then

- The row rank of A is the dimension of the span of the rows of A in $\mathbf{F}^{1,n}$.
- The column rank of A is the dimension of the span of the columns of A in $\mathbf{F}^{m,1}$.

Proposition 3.80. column-row factorization
Suppose A is an m-by-n matrix with column rank $c \geq 1$. Then there exist an m-by-c matrix C and a c-by-n matrix R such that

$$A = CR$$

Proposition 3.81. Suppose C is an m-by-c matrix and R is a c-by-n matrix, then

- For $k \in \{1, \ldots, n\}$, column k of CR is a linear combination of the columns of C, with the coefficients coming from column k of R.
- For $k \in \{1, \ldots, m\}$, row k of CR is a linear combination of the rows of R, with the coefficients coming from row k of C.

Proposition 3.82. Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then $\dim \operatorname{range} T$ equals the column rank of $\mathcal{M}(T)$.

Proposition 3.83. Suppose $A \in \mathbf{F}^{m,n}$. Then the row rank of A equals the column rank of A.

Remark.

$$\text{range } T = \text{range } T'$$

Definition 3.84. Rank
The rank of a matrix $A \in \mathbf{F}^{m,n}$ is the column rank of A.

4 Polynomials

Proposition 4.1. Suppose $a_0, \ldots, a_m \in \mathbf{F}$, then

$$\forall z \in \mathbf{F},\ a_0 + a_1 z + \cdots + a_m z^m = 0 \implies a_0 = \cdots = a_m = 0$$

Remark. This shows that the coefficients are uniquely determined, i.e. the same function is equivalent to the same coeffiecient.

Proposition 4.2. Suppose that $p, s \in \mathcal{P}(\mathbf{F})$ with $s \neq 0$. Then there exist unique polynomials $q, r \in \mathcal{P}(\mathbf{F})$ such that $\deg r < \deg s$ and

$$p = sq + r$$

Definition 4.3. Zero of a polynomial
A number $\lambda \in \mathbf{F}$ is called a zero (or root) of a polynomial $p \in \mathcal{P}(\mathbf{F})$ if

$$p(\lambda) = 0$$

Definition 4.4. Factor
A polynomial $s \in \mathcal{P}(\mathbf{F})$ is called a factor of $p \in \mathcal{P}(\mathbf{F})$ if there exists a polynomial $q \in \mathcal{P}(\mathbf{F})$ such that $p = sq$.

Proposition 4.5. Suppose $p \in \mathcal{P}(\mathbf{F})$ and $\lambda \in \mathbf{F}$.Then $p(\lambda) = 0$ iff there exists a polynomial $q \in \mathcal{P}(\mathbf{F})$ such that for every $z \in \mathbf{F}$

$$p(z) = (z - \lambda)q(z)$$

Proposition 4.6. Suppose $p \in \mathcal{P}(\mathbf{F})$ is a polynomial with degree $m \geq 0$. Then p has at most m distinct zeros in $\mathbf{F}$.

Theorem 4.7. Every nonconstant polynomial with complex coefficients has a zero.

Remark. This is the **Fundamental Theorem of Algebra**.

Proposition 4.8. Suppose $p \in \mathcal{P}(\mathbf{C})$ is a nonconstant polynomial, then p has a unique factorization (regardless of the order of the factors) of the form

$$p(z) = c(z - \lambda_1) \cdots (z - \lambda_m)$$

where $c, \lambda_1, \ldots, \lambda_m \in \mathbf{C}$.

Proposition 4.9. Suppose $p \in \mathcal{P}(\mathbf{C})$ is a polynomial with real coefficients. If $\lambda \in \mathbf{C}$ is a zero of p, then so is $\bar{\lambda}$

Remark. Just by taking complex conjugate of both side of the equation

$$a_0 + a_1 z + \cdots + a_m z^m = 0$$

To further prove that they appear the same number of times, we can use Proposition 4.5 to cancel each one of them and then prove it by induction.

Proposition 4.10. Suppose $p \in \mathcal{P}(\mathbf{R})$ is a nonconstant polynomial, then p has a unique factorization (regardless of the order of the factors) of the form

$$p(x) = c(x - \lambda_1) \cdots (x - \lambda_m)(x^2 + b_1 x + c_1) \cdots (x^2 + b_M x + c_M)$$

where $c, \lambda_1, \ldots, \lambda_m, b_1, \ldots, b_M, c_1, \ldots, c_M \in \mathbf{R}$, with $b_i^2 < 4c_i$ for each i.

5 Eigenvalues and Eigenvectors

5.1 Invariant Subspaces

Definition 5.1. Invariant subspace
Suppose $T \in \mathcal{L}(V)$. A subspace U of V is called invariant under T if

$$u \in U \implies T(u) \in U$$

Example 5.2. Suppose $T \in \mathcal{L}(V)$. Then the following special subspaces are invariant subspaces of V:

- $\{0\}$
- V
- null T
- range T

Definition 5.3. Eigenvalue
Suppose $T \in \mathcal{L}(V)$. $\lambda \in \mathbf{F}$ is called an eigenvalue if there exists $v \in V$ such that $v \neq 0$ and $T(v) = \lambda v$.

Proposition 5.4. Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, $\lambda \in \mathbf{F}$, then the following are equivalent:

- λ is an eigenvalue of T
- $T - \lambda I$ is not injective
- $T - \lambda I$ is not surjective
- $T - \lambda I$ is not invertible

Remark. The last three statements are equivalent by Proposition 3.44.

Definition 5.5. Eigenvector
Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbf{F}$ is an eigenvalue of T. A vector $v \in V$ is called an eigenvector of T corresponding to λ if $v \neq 0$ and $T(v) = \lambda v$.

Remark. A sufficient and neccessary condition would be $v \in \operatorname{null}(T - \lambda I)$.

Proposition 5.6. Let $T \in \mathcal{L}(V)$. Suppose $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of T and their corresponding eigenvectors are $v_1, \ldots, v_m$. Then $v_1, \ldots, v_m$ is linearly independent.

Corollary 5.7. Suppose V is finite-dimensional, then each operator on V has at most $\dim V$ distinct eigenvalues.

Definition 5.8. Restriction operator & quotient operator
Suppose $T \in \mathcal{L}(V)$ and U is a subspace of V invariant under T. Then

- The restriction operator $T|_U \in \mathcal{L}(U)$ is defined by

  $$T|_U(u) = T(u)$$

  for $u \in U$.

- The quotient operator $T/U \in \mathcal{L}(V/U)$ is defined by

  $$T/U(v + U) = T(v) + U$$

  for $v \in V$.

Definition 5.9. T^m
Suppose $T \in \mathcal{L}(V)$,

- For $m \in \mathbf{Z}^+$, T^m is defined by

  $$T^m = \underbrace{T \cdots T}_{m \text{ times}}$$

- T^0 is defined to be the identity operator I on V.

- If T is invertible with inverse T^{-1}, then T^{-m} is defined by

  $$T^{-m} = (T^{-1})^m$$

Definition 5.10. $p(T)$
Suppose $T \in \mathcal{L}(V)$ and $p \in \mathcal{P}(\mathbf{F})$ is a polynomial given by

$$p(z) = a_0 + a_1 z + \cdots + a_m z^m$$

for $z \in \mathbf{F}$.
Then $p(T)$ is the operator defined by

$$p(T) = a_0 I + a_1 T + \cdots + a_m T^m$$

Definition 5.11. Product of polynomials
Suppose $p, q \in \mathcal{P}(\mathbf{F})$, then $pq \in \mathcal{P}(\mathbf{F})$ is the polynomial defined by

$$(pq)(z) = p(z)q(z)$$

Proposition 5.12. Suppose $p, q \in \mathcal{P}(\mathbf{F})$ and $T \in \mathcal{L}(V)$. Then

$$(pq)(T) = p(T)q(T) = q(T)p(T)$$

Proposition 5.13. Suppose $T \in \mathcal{L}(V)$ and $p \in \mathcal{P}(\mathbf{F})$. Then null $p(T)$ and range $p(T)$ are invariant under T.

5.2 The Minimal Polynomial

Proposition 5.14. Every operator on a finite-dimensional, nonzero, complex vector space has an eigenvalue.

Remark. Think of the process how we find an eigenvalue, then prove it by the existance of complex root, i.e the Fundamental Theorem of Algebra.

Definition 5.15. Monic polynomial
A monic polynomial is a polynomial whose highest-degree coefficient equals 1.

Proposition 5.16. Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then there is a unique monic polynomial $p \in \mathcal{P}(\mathbf{F})$ of the smallest degree such that $p(T) = 0$. Furthermore, $\deg p \leq \dim V$.

Definition 5.17. Minimal polynomial
Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then the minimal polynomial of T is the unique monic polynomial $p \in \mathcal{P}(\mathbf{F})$ of smallest degree such that $p(T) = 0$.

Proposition 5.18. Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$, then

- The zeros of the minimal polynomial of T are the eigenvalues of T.
- If V is a complex vector space, then the minimal polynomial of T has the form
$$(z - \lambda_1) \cdots (z - \lambda_m)$$
where $\lambda_1, \ldots, \lambda_m$ is a list of all eigenvalues of T, possibly with repetitions.

Remark. Note that for the second point, we cannot obtain the exact minimal polynomial of T easily from its eigenvalues, but only the form. Equivalently, we only know that $z - \lambda$ is a factor of the minimal polynomial, but cannot determine the exact degree, as is meant by "possibly with repetitions".

Proposition 5.19. Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$ and $p \in \mathcal{P}(\mathbf{F})$. Then $q(T) = 0$ iff q is a polynomial multiple of the minimal polynomial of T.

Corollary 5.20. Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and U is a subspace of V that is invariant under T. Then the minimal polynomial of T is a polynomial multiple of the minimal polynomial of $T|_U$.

Proposition 5.21. Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then T is not invertible iff the constant term of the minimal polynomial of T is 0.

Remark. Consider Proposition 5.4 and Proposition 5.18.

Proposition 5.22. Suppose $\mathbf{F} = \mathbf{R}$ and V is finite-dimensional. Suppose also that $T \in \mathcal{L}(V)$ and $b, c \in \mathbf{R}$ with $b^2 < 4c$. Then $\dim \operatorname{null} (T^2 + bT + cI)$ is an even number.

Remark. Notice that $T^2 + bT + cI = 0$ implies T has no eigenvalue.

Proposition 5.23. Every operator on an odd-dimensional real vector space has an eigenvalue.

Remark. Analogy: Every odd-degree polynomial with real coefficients have a real root.

5.3 Upper-Triangular Matrices

Definition 5.24. Matrix of an operator
Suppose $T \in \mathcal{L}(V)$ and $v_1, \ldots, v_n$ is a basis of V. The matrix of T, denoted $\mathcal{M}(T)$, with respect to this basis is the n-by-n matrix defined by

$$\mathcal{M}(T) = \begin{pmatrix} A_{1,1} & \cdots & A_{1,n} \\ \vdots & & \vdots \\ A_{n,1} & \cdots & A_{n,n} \end{pmatrix}$$

where the entries $A_{i,j}$ are defined by

$$T(v_j) = A_{1,j}v_1 + \cdots + A_{n,j}v_n$$

We also use the notation $\mathcal{M}(T, (v_1, \ldots, v_n))$ to clearify the basis.

Definition 5.25. Diagonal of a matrix
The diagonal of a square matrix consists of the entries along the line from the upper left corner to the bottom right corner.

Definition 5.26. Upper-triangular matrix
A matrix is called upper triangular if all the entries below the diagonal equal 0.

Proposition 5.27. Suppose $T \in \mathcal{L}(V)$ and $v_1, \ldots, v_n$ is a basis of V. Then the following are equivalent.

- the matrix of T with respect to $v_1, \ldots, v_n$ is upper triangular.
- $T(v_i) \in \text{span}(v_1, \ldots, v_i)$ for each $i \in \{1, \ldots, n\}$
- $\text{span}(v_1, \ldots, v_i)$ is invariant under T for each $i \in \{1, \ldots, n\}$

Proposition 5.28. Suppose V is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$. Then T has an upper-triangular matrix with some basis of V.

Remark. We can consider the restriction operator $T|_U$ on a subspace U of V, then prove this proposition by induction.

Remark. Alternatively, we can consider the quotient operator T/U and we can also apply our induction hypothesis by Proposition 3.58.

Proposition 5.29. Suppose $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to some basis of V. Then T is invertible iff all the entries on the diagonal of that upper-triangular matrix are nonzero.

Proposition 5.30. Suppose $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to some basis of V. Then the eigenvalues of T are precisely the entries on the diagonal of that upper-triangular matrix.

Proposition 5.31. Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then T has an upper-triangular matrix with respect to some basis of V iff the minimal polynomial of T equals $(z - \lambda_1) \cdots (z - \lambda_m)$ for some $\lambda_1, \ldots, \lambda_m \in \mathbf{F}$.

5.4 Eigenspaces and Diagonal Matrices

Definition 5.32. Diagonal matrix
A diagonal matrix is a square matrix that is 0 everywhere except possibly along the diagonal.

Definition 5.33. Eigenspace
Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbf{F}$. The eigenspace of T corresponding to λ, denoted $E(\lambda, T)$, is defined by

$$E(\lambda, T) = \operatorname{null}(T - \lambda I)$$

In other words, $E(\lambda, T)$ is set of all eigenvectors of T corresponding to λ, along with the 0 vector.

Remark. This implies that λ is an eigenvalue of T iff $E(\lambda, T) \neq \{0\}$

Remark. If λ is an eigenvalue of T, then it implies $T|_{E(\lambda,T)}(v) = \lambda v$.

Proposition 5.34. Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Suppose also that $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of T. Then

$$E(\lambda_1, T) + \cdots + E(\lambda_m, T)$$

is a direct sum. Furthermore, we have

$$\dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T) \leq \dim V$$

Definition 5.35. Diagonalizable
An operator $T \in \mathcal{L}(V)$ is diagonalizable if the operator has a diagonal matrix with respect to some basis of V.

Proposition 5.36. Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Let $\lambda_1, \ldots, \lambda_m$ denote the distinct eigenvalues of T. Then the following are equivalent.

- T is diagonalizable
- V has a basis consisting of eigenvectors of T.
- there exists 1-dimensional subspaces $U_1, \ldots, U_n$ of V, each invariant under T, such that
$$V = U_1 \oplus \cdots \oplus U_n$$
- $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$
- $\dim V = \dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T)$

Proposition 5.37. If $T \in \mathcal{L}(V)$ has $\dim V$ distinct eigenvalues, then T is diagonalizable.

Remark. Note that this is not a neccessary condition for diagonalizability.

Proposition 5.38. Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Then T is diagonalizable iff the minimal polynomial of T equals $(z - \lambda_1) \cdots (z - \lambda_m)$ for some list of distinct numbers $\lambda_1, \ldots, \lambda_m \in \mathbf{F}$.

Remark. That means, T is diagonalizable iff its minimal polynomial has no repeated roots.

Corollary 5.39. Suppose $T \in \mathcal{L}(V)$ is diagonalizable and U is a subspace of V that is invariant under T. Then $T|_U$ is a diagonalizable operator on U.

Definition 5.40. Gershgorin disks
Suppose $T \in \mathcal{L}(V)$ and $v_1, \ldots, v_n$ is a basis of V. Let A denote the matrix of T with respect to this basis. A Gershgorin disk of T with respect to this basis is a set of the form

$$\left\{ z \in \mathbf{F} \;\middle|\; |z - A_{i,i}| \leq \sum_{j=1, j \neq i}^{n} |A_{i,j}| \right\}$$

where $i \in \{1, \ldots, n\}$.

Theorem 5.41. Gershgorin disk theorem
Suppose $T \in \mathcal{L}(V)$ and $v_1, \ldots, v_n$ is a basis of V. Then each eigenvalue of T is contained in some Gershgorin disk of T with respect to the basis $v_1, \ldots, v_n$.

5.5 Commuting Operators

Definition 5.42. Commute

- Two operators S and T on the same vector space commute if $ST = TS$.
- Two square matrix A and B of the same size commute if $AB = BA$.

Remark. Note that the identity operator I commutes with every operator on V.

Proposition 5.43. Suppose $S, T \in \mathcal{L}(V)$. Then S and T commute iff $\mathcal{M}(S)$ and $\mathcal{M}(T)$ commute.

Proposition 5.44. Suppose $S, T \in \mathcal{L}(V)$ commute and $\lambda \in \mathbf{F}$, then $E(\lambda, S)$ is invariant under T.

Proposition 5.45. Two diagonalizable operators on the same vector space possibly have diagonal matrices with respect to the same basis iff the two operators commute.

Remark. This does not mean they have exactly the same set of eigenvectors, just by considering the identity operator I, which commutes with every operator.
This is because the proof just makes sure that they have a list of common eigenvectors with length $\dim V$. More explicitly, the proof shows that they have common eigenvectors in every eigenspace, which may consist of linear combinations of multiple eigenvectors.

For example, if λ_1, λ_2 are eigenvectors of S with the same eigenvalue, and λ_3, which is a linear combination of λ_1, λ_2, is an eigenvector of T, which has the same eigenvalue with λ_4, which is not a linear combination of λ_1, λ_2. Therefore, $\text{span}(\lambda_1, \lambda_2)$ are eigenvectors of S, while $\text{span}(\lambda_3, \lambda_4)$ are eigenvectors of T, which is not exactly the same.

Proposition 5.46. Every pair of commuting operators on a finite-dimensional nonzero complex vector space has a common eigenvector.

Remark. Complex vector space only makes sure the existance of eigenvector, as stated in Proposition 5.14.

Proposition 5.47. Suppose V is a finite-dimensional complex vector space and S, T are commuting operators on V. Then there is a basis of V with respect to which both S and T have upper-triangular matrices.

Remark. An extension of Proposition 5.28.

Proposition 5.48. Suppose V is a finite-dimensional complex vector space and S, T are commuting operators on V. Then

- every eigenvalue of $S + T$ is an eigenvalue of S plus an eigenvalue of T
- every eigenvalue of ST is an eigenvalue of S times an eigenvalue of T

Remark. Just by consider the upper-triangular form of S and T.

6 Inner Product Spaces

6.1 Inner Products and Norms

Definition 6.1. Dot Product
For $x, y \in \mathbf{R}^n$, the dot product of x and y, denoted $x \cdot y$, is defined by

$$x \cdot y = x_1 y_1 + \cdots + x_n y_n$$

where $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$.

Definition 6.2. Inner Product
An inner product on V is a function that takes each ordered pair $(u, v) \in V \times V$ to a number $\langle u, v \rangle \in \mathbf{F}$ and has the following properties:

- **positivity**
$$\langle v, v \rangle \geq 0$$
- **definiteness**
$$\langle v, v \rangle = 0 \iff v = 0$$
- **additivity in first slot**
$$\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$$
- **homogeneity in first slot**
$$\langle \lambda u, v \rangle = \lambda \langle u, v \rangle$$
- **conjugate symmetry**
$$\langle u, v \rangle = \overline{\langle v, u \rangle}$$

Remark. The positivity implies that $\langle v, v \rangle \in \mathbf{R}$.

Example 6.3.

- The Euclidean inner product on $\mathbf{F}^n$ is defined by
$$\langle (w_1, \ldots, w_n), (z_1, \ldots, z_n) \rangle = w_1 \overline{z_1} + \cdots + w_n \overline{z_n}$$
- let $c_1, \ldots, c_n > 0$, then we can define an inner product on $\mathbf{F}^n$ by
$$\langle (w_1, \ldots, w_n), (z_1, \ldots, z_n) \rangle = c_1 w_1 \overline{z_1} + \cdots + c_n w_n \overline{z_n}$$
- We can define the an inner product on the vector space of continous real-valued functions on the interval $[-1, 1]$ by
$$\langle f, g \rangle = \int_{-1}^{1} f(x) g(x) \, \mathrm{d}x$$
- An inner product on $\mathcal{P}(\mathbf{R})$ can be defined by
$$\langle p, q \rangle = \int_{0}^{\infty} p(x) q(x) e^{-x} \, \mathrm{d}x$$

Definition 6.4. Inner product space
An inner product space is a vector space along with an inner product on V.

Proposition 6.5.

- For each fixed $u \in V$, the function that takes v to $\langle v, u \rangle$ is a linear map from V to $\mathbf{F}$.
- $\langle 0, u \rangle = \langle u, 0 \rangle = 0$ for all $u \in V$.
- $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle$ for all $u, v, w \in V$
- $\langle u, \lambda v \rangle = \bar{\lambda} \langle u, v \rangle$ for all $\lambda \in \mathbf{F}$ and $u, v \in V$.

Definition 6.6. Norm
For $v \in V$, the norm of v, denoted $\|v\|$, is defined by

$$\|v\| = \sqrt{\langle v, v \rangle}$$

Proposition 6.7. Suppose $v \in V$, then

- $\|v\| = 0 \iff v = 0$
- $\|\lambda v\| = |\lambda| \, \|v\|$ for all $\lambda \in \mathbf{F}$

Definition 6.8. Orthogonal
Two vectors $u, v \in V$ are called orthogonal if $\langle u, v \rangle = 0$

Proposition 6.9.

- 0 is orthogonal to every vector in V.
- 0 is the only vector in V that is orthogonal to itself.

Theorem 6.10. Pythagorean Theorem
Suppose u and v are orthogonal vectors in V. Then

$$\|u+v\|^2 = \|u\|^2 + \|v\|^2$$

Proposition 6.11. Orthogonal decomposition
Suppose $u, v \in V$ with $v \neq 0$. Let $c = \dfrac{\langle u, v\rangle}{\|v\|}$ and $w = u - \dfrac{\langle u, v\rangle}{\|v\|}v$. Then

$$\langle w, v\rangle = 0 \quad \text{and} \quad u = cv + w$$

Theorem 6.12. Cauchy–Schwarz Inequality
Suppose $u, v \in V$. Then

$$|\langle u, v\rangle| \leq \|u\|\,\|v\|$$

where equality holds iff one of u and v is a scalar multiple of the other.

Remark. Applied the Euclidean inner product on $\mathbf{R}^n$, the inequality becomes

$$(\sum_{i=1}^{n} a_i b_i)^2 \leq (\sum_{i=1}^{n} a_i^2)(\sum_{i=1}^{n} b_i^2)$$

which we are rather familiar with.

Theorem 6.13. Triangle Inequality
Supppose $u, v \in V$, then

$$\|u+v\| \leq \|u\| + \|v\|$$

where equality holds iff one of u and v is a nonnegative multiple of the other.

Theorem 6.14. Parallelogram Equality
Suppose $u, v \in V$. Then

$$\|u+v\|^2 + \|u-v\|^2 = 2(\|u\|^2 + \|v\|^2)$$

6.2 Orthonormal Bases

Definition 6.15. Orthonormal
A list of vectors are called orthonormal if each vector in the list has norm 1 and is orthogonal to all the other vectors in the list. In other words, a list $e_1, \ldots, e_m$ is orthonormal if

$$\langle e_i, e_j \rangle = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

Proposition 6.16. If $e_1, \ldots, e_m$ is an orthonormal list of vectors in V, then

$$\|a_1 e_1 + \cdots + a_m e_m\|^2 = |a_1|^2 + \cdots + |a_m|^2$$

for all $a_1, \ldots, a_m \in \mathbf{F}$.

Remark. This implies that any orthonormal list of vectors is linearly independent.

Definition 6.17. Orthonormal basis
An orthonormal basis of V is an orthonormal list of vectors in V that is also a basis of V.

Proposition 6.18. Every orthonormal list of vectors in V with length $\dim V$ is an orthonormal basis of V.

Proposition 6.19. Suppose $e_1, \ldots, e_n$ is an orthonormal basis of V and $u, v \in V$. Then

- $v = \langle v, e_1 \rangle e_1 + \cdots + \langle v, e_n \rangle e_n$
- $\|v\|^2 = |\langle v, e_1 \rangle|^2 + \cdots + |\langle v, e_m \rangle|^2$
- $\langle u, v \rangle = \langle u, e_1 \rangle \overline{\langle v, e_1 \rangle} + \cdots + \langle u, e_n \rangle \overline{\langle v, e_n \rangle}$

Theorem 6.20. Gram–Schmidt Procedure
Suppose $v_1, \ldots, v_m$ is a linearly independent list of vectors in V. Let $e_1 = v_1 / \|v_1\|$. For $i = 2, \ldots, m$, defined e_i inductively by

$$e_i = \frac{v_i - \langle v_i, e_1 \rangle e_1 - \cdots - \langle v_i, e_{j-1} \rangle e_{i-1}}{\|v_i - \langle v_i, e_1 \rangle e_1 - \cdots - \langle v_i, e_{j-1} \rangle e_{i-1}\|}$$

Then $e_1, \ldots, e_m$ is an orthonormal list of vectors in V such that

$$\operatorname{span}(v_1, \ldots, v_i) = \operatorname{span}(e_1, \ldots, e_i)$$

for $i = 1, \ldots, m$

Corollary 6.21. Every finite-dimensional inner product space has an orthonormal basis.

Proposition 6.22. Suppose V is finite-dimensional. Then every orthonormal list of vectors in V can be extended to an orthonormal basis of V.

Remark. Referring to Proposition 2.16.

Proposition 6.23. Suppose $T \in \mathcal{L}(V)$. If T has an upper-triangular matrix with respect to some basis of V, then T has an upper-triangular matrix with respect to some **orthonormal** basis of V.
Equivalently, T has an upper-triangular matrix with respect to some orthonormal basis of V iff the minimal polynomial of T equals $(z - \lambda_1) \cdots (z - \lambda_m)$ for some $\lambda_1, \ldots, \lambda_m \in \mathbf{F}$.

Corollary 6.24. Schur's Theorem
Suppose V is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$ Then T has an upper-triangular matrix with respect to some orthonormal basis of V.

Remark. Referring to Proposition 5.28.

Theorem 6.25. Riesz Representation Theorem
Suppose V is finite-dimensional and φ is a linear functional on V. Then there is a unique vector $u \in V$ such that

$$\varphi(v) = \langle v, u \rangle$$

for all $v \in V$.

Remark. Recall that a linear functional on V is the linear map from V to $\mathbf{F}$ as defined in Definition 3.61. Thus for fixed $u \in V$, the inner product $\langle v, u \rangle$ is a linear functional.
Furthermore, we denote $\varphi_u(v) = \langle v, u \rangle$ for fixed u. Then the function $u \mapsto \varphi_u$ from V to V' is surjective.

Remark. Consider

$$\begin{aligned}
\varphi(v) &= \varphi(\langle v, e_1 \rangle e_1 + \cdots + \langle v, e_n \rangle e_n) \\
&= \langle v, e_1 \rangle \varphi(e_1) + \cdots + \langle v, e_n \rangle \varphi(e_n) \\
&= \langle v, \overline{\varphi(e_1)} e_1 + \cdots + \overline{\varphi(e_n)} e_n \rangle
\end{aligned}$$

6.3 Orthogonal Complements and Minimization Problems

Definition 6.26. Orthogonal complement
If U is a subset of V, then the orthogonal complement of U, denoted $U^\perp$, is the set of all vectors in V that are orthogonal to every vector in U, i.e.

$$U^\perp = \{v \in V \mid \langle v, u\rangle = 0, \forall u \in U\}$$

Proposition 6.27. Basic properties of orthogonal complements

- If U is the subset of V, then $U^\perp$ is the subspace of V.
- $\{0\}^\perp = V$
- $V^\perp = \{0\}$
- If U is a subset of V, then $U \cap U^\perp \subset \{0\}$
- If U and W are subsets of V with $U \subset W$, then $W^\perp \subset U^\perp$

Proposition 6.28. Direct sum decomposition
Suppose U is a finite-dimensional subspace of V. Then

$$V = U \oplus U^\perp$$

Proposition 6.29. Suppose V is finite-dimensional and U is a subspace of V. Then

$$\dim U^\perp = \dim V - \dim U$$

Proposition 6.30. Suppose U is a finite-dimensional subspace of V, then

$$U = (U^\perp)^\perp$$

Definition 6.31. Orthogonal projection
Suppose U is a finite-dimensional subspace of V. The orthogonal projection of V onto U is the operator $P_U \in \mathcal{L}(V)$ defined as:
For $v \in V$ and $v = u + w$, where $u \in U$ and $w \in U^\perp$, then $P_U(v) = u$

Proposition 6.32. Properties of P_U
Suppose U is a finite-dimensional subspace of V and $v \in V$, then

- $P_U \in \mathcal{L}(V)$
- $P_U(u) = u$ for all $u \in U$
- $P_U(w) = 0$ for all $w \in U^\perp$
- range $P_U = U$
- null $P_U = U^\perp$
- $v - P_U v \in U^\perp$
- $P_U^2 = P_U$
- $\|P_U(v)\| \leq \|v\|$
- $\langle P_U(v), v\rangle = \|P_U(v)\|^2$
- For all orthonormal basis $e_1, \ldots, e_m$ of U,

$$P_U(v) = \langle v, e_1\rangle e_1 + \cdots + \langle v, e_m\rangle e_m$$

Proposition 6.33. Suppose U is a finite-fimensional subspace of V, $v \in V$ and $u \in U$. Then

$$\|v - P_U(v)\| \leq \|v - u\|$$

where the equility holds iff $u = P_U(v)$.

6.4 Pseudoinverse

Proposition 6.34. Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Then $T|_{(\text{null } T)^\perp}$ is injective and range $T|_{(\text{null } T)^\perp} = \text{range } T$.

Remark.

$$Tu = Tu' \implies u - u' \in \text{null } T \implies \langle u - u', u - u'\rangle = 0 \implies u = u'$$

Definition 6.35. Pseudoinverse
Suppose that V is finite-dimensional and $T \in \mathcal{L}(V, W)$. The pseudoinverse $T^\dagger \in \mathcal{L}(W, V)$ is the linear map from W to V defined by

$$T^\dagger w = (T|_{(\text{null } T)^\perp})^{-1} P_{\text{range } T} w$$

Proposition 6.36. Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Then

- If T is invertible, then $T^\dagger = T^{-1}$
- $TT^\dagger = P_{\text{range } T}$
- $T^\dagger T = P_{(\text{null } T)^\perp}$

Proposition 6.37. Suppose V is finite-dimensional, $T \in \mathcal{L}(V, W)$, $b \in W$. Then

- If $x \in V$, then
$$\left\|T(T^\dagger b) - b\right\| \leq \|Tx - b\|$$
where equality holds iff $x \in T^\dagger b + \text{null } T$
- If $x \in T^\dagger b + \text{null } T$, then
$$\left\|T^\dagger b\right\| \leq \|x\|$$
where equality holds iff $x = T^\dagger b$

Remark. Consider $Tx - b = (Tx - T(T^\dagger b)) + (T(T^\dagger b) - b)$, where $Tx - T(T^\dagger b) \in \text{range } T$ and $T(T^\dagger b) - b \in (\text{range } T)^\perp$.
Consider $x = (x - T^\dagger b) + T^\dagger b$, where $x - T^\dagger b \in \text{null } T$ and $T^\dagger b \in (\text{null } T)^\perp$

Remark. This proposition shows that the pseudoinverse is the **best fit** to the equation $Tx = b$.

7 Operators on Inner Product Spaces

Note that we assume V and W to be nonzero finite-dimensional inner product spaces over $\mathbf{F}$ in this section.

7.1 Self-Adjoint and Normal Operators

Definition 7.1. Adjoint
Suppose $T \in \mathcal{L}(V, W)$. The adjoint of T is the function $T^* : W \to V$ such that

$$\langle Tv, w \rangle = \langle v, T^* w \rangle$$

for all $v \in V$, $w \in W$

Remark. If we fix w, then $v \mapsto \langle Tv, w \rangle$ is a linear functional on V. By Riesz Representation Theorem, there is unique $u \in V$ such that

$$\langle Tv, w \rangle = \langle v, u \rangle$$

thus we define $T^* w = u$ here.

Proposition 7.2. If $F \in \mathcal{L}(V, W)$, then $T^* \in \mathcal{L}(W, V)$.

Proposition 7.3. Properties of the adjoint
Suppose $T \in \mathcal{L}(V, W)$, then

- $(S + T)^* = S^* + T^*$ for all $S \in \mathcal{L}(V, W)$
- $(\lambda T)^* = \bar{\lambda} T^*$ for all $\lambda \in \mathbf{F}$
- $(T^*)^* = T$
- $(ST)^* = T^* S^*$ for all $S \in \mathcal{L}(W, U)$
- $I^* = I$ where I is the identity operator on V
- If T is invertible, then T^* is invertible and $(T^*)^{-1} = (T^{-1})^*$

Remark. If $\mathbf{F} = \mathbf{R}$, then the map $T \to T^*$ is linear from the first two properties.

Proposition 7.4. Suppose $T \in \mathcal{L}(V, W)$, then

- null $T^* = (\text{range } T)^\perp$
- range $T^* = (\text{null } T)^\perp$
- null $T = (\text{range } T^*)^\perp$
- range $T = (\text{range } T^*)^\perp$

Definition 7.5. Conjugate transpose
The conjugate transpose of an m-by-n matrix A is the n-by-m matrix A^* obtained by interchanging the rows and colomns and then taking the complex conjugate of each entry, i.e.

$$(A^*)_{i,j} = \overline{A_{j,i}}$$

for all $j = 1, \ldots, n$ and all $k = 1, \ldots, m$.

Proposition 7.6. Let $T \in \mathcal{L}(V, W)$. Suppose $e_1, \ldots, e_n$ is an **orthonormal** basis of V and $f_1, \ldots, f_m$ is an orthonormal basis of W, then $\mathcal{M}(T^*, (f_1, \ldots, f_m), (e_1, \ldots, e_n))$ is the conjugate transpose of $\mathcal{M}(T, (e_1, \ldots, e_n), (f_1, \ldots, f_m))$, i.e.

$$\mathcal{M}(T^*) = (\mathcal{M}(T))^*$$

Remark. There is remarkable relationship and analogy between inner product space and ordinary vector space:

- V & V': consider function $u \mapsto \varphi_u(v) = \langle v, u \rangle$ from V to V', as mensioned in Riesz Representation Theorem.
- U^0 & $U^\perp$:
  - $U^0 = \{\varphi \in V' \mid \varphi(u) = 0\}$ corresponds to $U^\perp = \{v \in V \mid \langle v, u \rangle = 0\}$.
  - $\dim U^0 = \dim U^\perp$, as proved in Proposition 3.71 and Proposition 6.29.
- T^* & T':
  - $T^* : W \to V$ corresponds to $T' : W' \to V'$, both like representing the basis of W as linear combination of the basis of V.
  - Furthermore, $\varphi_{T^*w}(v) = \langle v, T^*w \rangle = \langle Tv, w \rangle = \varphi_w(Tv) = T'(\varphi_w)(v)$
  - null $T^* = (\text{range } T)^\perp$ corresponds to null $T' = (\text{range } T)^0$, as proved in Proposition 7.4 and Proposition 3.72.
  - range $T^* = (\text{null } T)^\perp$ corresponds to range $T' = (\text{null } T)^0$, as proved in Proposition 7.4 and Proposition 3.74.

Definition 7.7. Self-adjoint
An operator $T \in \mathcal{L}(V)$ is called self-adjoint if $T = T^*$.

Remark. Suppose $e_1, \ldots, e_n$ is an orthonormal basis of V, then T is self-adjoint iff $\mathcal{M}(T) = (\mathcal{M}(T))^*$.

Proposition 7.8. Every eigenvalue of a self-adjoint operator is real.

Proposition 7.9. Suppose V is a complex inner product space and $T \in \mathcal{L}(V)$. Then

$$\forall v \in V,\ \langle v, T(v)\rangle = 0 \iff T = 0$$

Remark. Consider

$$\langle Tu, w\rangle = \frac{\langle T(u+w), u+w\rangle - \langle T(u-w), u-w\rangle}{4} + \frac{\langle T(u+iw), u+iw\rangle - \langle T(u-iw), u-iw\rangle}{4} i$$

Proposition 7.10. Suppose V is a complex inner product space and $T \in \mathcal{L}(V)$. Then T is self-adjoint $\iff \forall v \in V,\ \langle Tv, v\rangle \in \mathbf{R}$.

Proposition 7.11. Suppose T is a self-ajoint operator on V. Then

$$\forall v \in V,\ \langle Tv, v\rangle = 0 \iff T = 0$$

Definition 7.12. Normal
An operator on an inner product space is called normal if it commutes with its adjoint. In other words, $T \in \mathcal{L}(V)$ is normal if

$$TT^* = T^*T$$

Proposition 7.13. Suppose $T \in \mathcal{L}(V)$, then T is normal iff for all $v \in V$,

$$\|Tv\| = \|T^*v\|$$

Proposition 7.14. Suppose $T \in \mathcal{L}(V)$ is normal. Then

- null T = null T^*

- range T = range T^*
- V = null $T \oplus$ range T
- $T - \lambda I$ is normal for every $\lambda \in \mathbf{F}$
- $Tv = \lambda v \iff T^*v = \bar{\lambda}v$ for $v \in V$ and $\lambda \in \mathbf{F}$.

Proposition 7.15. Suppose $T \in \mathcal{L}(V)$ is normal. Then eigenvectors of T corresponding to distinct eigenvalues are orthogonal.

Remark. Consider

$$\begin{aligned}(\lambda_1 - \lambda_2)\langle v_1, v_2\rangle &= \langle \lambda_1 v_1, v_2\rangle - \langle v_1, \bar{\lambda_2} v_2\rangle \\ &= \langle Tv_1, v_2\rangle - \langle v_1, T^*v_2\rangle \\ &= 0\end{aligned}$$

Proposition 7.16. Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Then T is normal iff there exist commuting self-adjoint operators A and B such that $T = A + iB$.

Remark. Consider

$$A = \frac{T + T^*}{2} \quad \text{and} \quad B = \frac{T - T^*}{2i}$$

Remark. This provides an analogy between $\mathcal{L}(V)$ and $\mathbf{C}$.

7.2 Spectral Theorem

Proposition 7.17. Suppose $T \in \mathcal{L}(V)$ is self-adjoint and $b, c \in \mathbf{R}$ are such that $b^2 < 4c$. Then

$$T^2 + bT + cI$$

is an invertible operator.

Remark. There is analogy between real numbers and self-ajoint operators.

Proposition 7.18. Suppose $T \in \mathcal{L}(V)$ is self-adjoint. Then the minimal polynomial of T equals $(z - \lambda_1) \cdots (z - \lambda_m)$ for some $\lambda_1, \ldots, \lambda_m \in \mathbf{R}$.

Remark. For $\mathbf{F} = \mathbf{C}$ it's trivial.
For $\mathbf{F} = \mathbf{R}$, $T^2+bT+cI$ is invertible means that null $(T^2+bT+cI) = \{0\}$, meaning that $\nexists v \in V$ such that $(T^2+bT+cI)v = 0$, meaning that $(T-\lambda_1 I)\cdots(T-\lambda_m I)(T^2+bT+cI) = 0 \implies (T-\lambda_1 I)\cdots(T-\lambda_m I) = 0$. In this way, we can cancel out all the terms like $T^2+bT+cI$.

Remark. A more direct method is to multiply the inverse of $T^2+bT+cI$ (as it's invertible) on both sides to obtain $(T-\lambda_1 I)\cdots(T-\lambda_m I)(T^2+bT+cI) = 0 \implies (T-\lambda_1 I)\cdots(T-\lambda_m I) = 0$.

Remark. This implies every self-ajoint operator has an eigenvalue.

Remark. This implies every self-ajoint operator has an upper-triangular matrix with respect to some basis of V.

Theorem 7.19. Real spectral theorem
Suppose $\mathbf{F} = \mathbf{R}$ and $T \in \mathcal{L}(V)$. Then the following are equivalent

- T is self-ajoint.
- T has a diagonal matrix with respect to some orthonormal basis of V.
- V has an orthonormal basis consisting of eigenvectors of T.

Remark. Proposition 7.18 indicates that the minimal polynomial of any self-adjoint operator has the form of $(z-\lambda_1)\cdots(z-\lambda_m)$, then Proposition 6.23 shows that T has an upper-triangular matrix with respect to some orthonormal basis of V. Then by considering the definition of self-adjoint, we can directly prove the upper-triangular matrix is a diagonal matrix.

Remark. Self-ajoint operators are diagonalizable further implies that all λ_i in Proposition 7.18 are distinct.

Theorem 7.20. Complex spectral theorem
Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Then the following are equivalent

- T is normal
- T has a diagonal matrix with respect to some orthonormal basis of V.
- V has an orthonormal basis consisting of eigenvectors of T.

Remark. Similarly, Schur's Theorem shows that any operator on complex vector space has an upper-triangular matrix with respect to some othonormal basis. Then considering T is normal, we can actually prove that the upper-triangular matrix is diagonal, by expressing $\|Te_i\|$ and $\|T^*e_i\|$ and by Proposition 7.13.

7.3 Positive Operators

Definition 7.21. Positive operator
An operator $T \in \mathcal{L}(V)$ is called positive if T is self-adjoint and

$$\langle Tv, v \rangle \geq 0$$

for all $v \in V$.

Remark. If V is complex vector space, then by Proposition 7.10, $\langle Tv, v \rangle \geq 0 \implies$ T is self-adjoint.

Example 7.22.

- If U is a subspace of V, then the othogonal projection P_U is a positive operator.
- If $T \in \mathcal{L}(V)$ is self-adjoint and $b, c \in \mathbf{R}$ such that $b^2 < 4c$, then $T^2 + bT + cI$ is a positive operator.

Definition 7.23. Square root
An operator R is called a square root of an operator T if $R^2 = T$.

Proposition 7.24. Suppose $T \in \mathcal{L}(V)$. Then the following are equivalent

- T is a positive operator.
- T is self-adjoint and all eigenvalues of T are nonnegative.
- With respect to some orthonormal basis of V, the matrix of T is a diagonal matrix with only nonnegative numbers on the diagonal.
- T has a positive square root.
- T has a self-adjoint square root.
- $T = R^*R$ for some $R \in \mathcal{L}(V)$.

Remark. Here **"nonnegative"** means **real and nonnegative**.

Remark. There is an anology between characteristics of operators and complex numbers: For $z \in \mathbf{C}$, z is a nonnegative complex number (i.e. z is nonnegative real number) is equivalent to:

- z has a nonnegative square root.
- z has a real square root. (Recall that there's an anology between real numbers and self-adjoint operators, as mensioned in Proposition 7.17)
- $\exists w \in \mathbf{C}$, $z = \bar{w}w$.

These three neccessary and sufficient conditions correspond to the last three points in Proposition 7.24.

Proposition 7.25. Every positive operator on V has a unique positive square root.

Notation 7.26. $\sqrt{T}$
For T a positive operator, $\sqrt{T}$ denotes the unique positve square root of T.

7.4 Isometries, Unitary Operators, and Matrix Factorization

Definition 7.27. Isometry
A linear map $S \in \mathcal{L}(V, W)$ is called an isometry if

$$\|Sv\| = \|v\|$$

for all $v \in V$. In other words, an isometry is a linear map that preserves norms.

Remark. Easy to prove that every isometry is injective.

Proposition 7.28. Suppose $S \in \mathcal{L}(V, W)$. Suppose $e_1, \ldots, e_n$ is an orthonormal basis of V and $f_1, \ldots, f_m$ is an orthonormal basis of W. Then the following are equivalent.

- S is an isometry.
- $S^*S = I$
- $\langle Su, Sv \rangle = \langle u, v \rangle$ for all $u, v \in V$
- $Se_1, \ldots, Se_n$ is an orthonormal list in W
- The columns of $\mathcal{M}(S, (e_1, \ldots, e_n), (f_1, \ldots, f_m))$ form an orthonormal list in $\mathbf{F}^m$ with respect to the Euclidean inner product.

Definition 7.29. Unitary operator
An operator $S \in \mathcal{L}(V)$ is called unitary if S is an invertible isometry.

Remark. Note that every isometry is injective, as mensioned above. We also know that for all operators on finite-dimensional vector spaces, injectivity is equivalent to invertibility. Thus here invertible means V is finite-dimensional.

Proposition 7.30. Suppose $S \in \mathcal{L}(V)$. Suppose $e_1, \ldots, e_n$ is an orthonormal basis of V. Then the following are equivalent

- S is a unitary operator
- $SS^* = S^*S = I$
- S is invertible and $S^{-1} = S^*$
- $Se_1, \ldots, Se_n$ is an orthonormal basis of V
- The rows of $\mathcal{M}(S, (e_1, \ldots, e_n))$ form an orthonormal list in $\mathbf{F}^n$ with respect to the Euclidean inner product.
- S^* is a unitary operator.

Remark. Suppose λ is an eigenvalue of a unitary operator. Then $|\lambda| = 1$.

Proposition 7.31. Suppose $\mathbf{F} = \mathbf{C}$ and $S \in \mathcal{L}(V)$. Then the following are equivalent.

- S is a unitary operator.
- There is an orthonormal basis V consisting of eigenvalues of S whose corresponding eigenvalues all have absolute value 1.

Definition 7.32. Unitary matrix
An n-by-n matrix is called unitary if its columns form an orthonormal list in $\mathbf{F}^n$.

Proposition 7.33. Suppose Q is an n-by-n matrix. Then the following are equivalent.

- Q is a unitary matrix

- The rows of Q forms an orthonormal list in $\mathbf{F}^n$
- $\|Qv\| = \|v\|$ for every $v \in \mathbf{F}^n$
- $Q^*Q = QQ^* = I$.

Proposition 7.34. QR factorization
Suppose A is a square matrix with linearly independent columns. Then there exists unique matrices Q and R such that Q is unitary, R is upper-triangular with only positive numbers on its diagonal, and that

$$A = QR$$

Remark. QR factorization can be used to solve the equation system rapidly since the inverse of Q is Q^*, which is the conjugate transpose of Q.

Proposition 7.35. A self-adjoint operator $T \in \mathcal{L}(V)$ is a positive invertible operator iff $\langle Tv, v\rangle > 0$ for every nonzero $v \in V$.

Definition 7.36. Positive definite
A matrix $B \in \mathbf{F}^{n,n}$ is called positive difinite if $B^* = B$ and

$$\langle Bx, x\rangle > 0$$

for every nonzero $x \in \mathbf{F}^n$.

Proposition 7.37. Cholesky factorization
Suppose B is a positive definite matrix. Then there exists a unique upper-triangular matrix R with only positive numbers on its diagonal such that

$$B = R^*R$$

Remark. Conside Proposition 7.24, there is invertible A such that $B = A^*A$, then make QR factorization to A, obtaining

$$B = R^*Q^*QR = R^*R$$

7.5 Singular Value Decomposition

Proposition 7.38. Suppose $T \in \mathcal{L}(V, W)$, then

- T^*T is a positive operator on V
- null T^*T = null T
- range T^*T = range T^*
- dim range T = dim range T^* = dim range T^*T

Definition 7.39. Singular values
Suppose $T \in \mathcal{L}(V, W)$. The singular values of T are the nonnegative square roots of the eigenvalues of T^*T, listed in decreasing order, each included as many times as the dimensions of the corresponding eigenspaces of T^*T.

Proposition 7.40. Suppose $T \in \mathcal{L}(V, W)$. Then

- T is injective $\iff$ 0 is not a singular value of T
- the number of positive sigular values of T equals dim range T.
- T is surjective $\iff$ number of singular values of T equals $\dim W$.

Proposition 7.41. Suppose $S \in \mathcal{L}(V, W)$. Then S is an isometry iff all sigular values of S equal 1.

Proposition 7.42. Singular value decomposition (SVD)
Suppose $T \in \mathcal{L}(V, W)$ and the positive singular values for T are $s_1, \ldots, s_m$. Then there exist orthonormal lists $e_1, \ldots, e_m$ in V and $f_1, \ldots, f_m$ in W such that

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_m\langle v, e_m\rangle f_m$$

for every $v \in V$.

Remark. Here

$$Te_k = \begin{cases} s_k f_k & 1 \le k \le m \\ 0 & m < k \le \dim V \end{cases}$$

Then, we can extend the orthonormal lists $(e_1, \dots, e_m)$ and $(f_1, \dots, f_m)$ to orthonormal bases $(e_1, \dots, e_{\dim V})$ and $(f_1, \dots, f_{\dim W})$ by Proposition 6.22. Therefore, wtih respect to these bases, the matrix of T has the simple form

$$\mathcal{M}(T)_{i,j} = \begin{cases} s_k & 1 \le i = j \le m \\ 0 & \text{otherwise} \end{cases}$$

Hence we can claim that every operator has a "diagonal matrix" with respect to appropriate orthonormal bases.

Remark. Note that this does not mean that every operator is "diagonalizable" since here the orthonormal bases of the domain and range can be distinct, even when the domain equals the range.

Proposition 7.43. SVD of adjoint and pseudoinverse
Suppose $T \in \mathcal{L}(V, W)$ and positive singular values of T are $s_1, \dots, s_m$. Suppose $e_1, \dots, e_m$ and $f_1, \dots, f_m$ are orthonormal lists in V and W such that

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_m\langle v, e_m\rangle f_m$$

for every $v \in V$, then we have

$$T^*w = s_1\langle w, f_1\rangle e_1 + \cdots + s_m\langle w, f_m\rangle e_m$$

and

$$T^\dagger w = \frac{\langle w, f_1\rangle}{s_1} e_1 + \cdots + \frac{\langle w, f_m\rangle}{s_m} e_m$$

for every $w \in W$.

Proposition 7.44. Suppose A is an M-by-n matrix of rank $m \ge 1$. Then there exist an M-by-m matrix B with orthonormal columns, an m-by-m diagonal matrix D with positive numbers on the diagonal, and an n-by-m matrix C with orthonormal columns such that

$$A = BDC^*$$

Remark.

- The columns of B are $f_1, \dots, f_m$
- The diagonal entries of D are $s_1, \dots, s_m$
- The columns of C are $e_1, \dots, e_m$

7.6 Consequences of Singular Value Decomposition

Proposition 7.45. Suppose $T \in \mathcal{L}(V, W)$ and s_1 be the largest singular value of T. Then

$$\|Tv\| \leq s_1 \|v\|$$

for all $v \in V$

Definition 7.46. Norm of a linear map
Suppose $T \in \mathcal{L}(V, W)$, then the norm of T, denoted by $\|T\|$, is defined by

$$\|T\| = \max \{\|Tv\| \mid v \in V, \|v\| \leq 1\}$$

Remark. We know from Proposition 7.43 that $\|Tv\| \leq s_1$ if $\|v\| \leq 1$. We know also that $\|Te_1\| = \|s_1 f_1\| = s_1$. Thus $\|T\|$ is actually the largest singular value of T.

Remark. Under such definition, we have

$$\|Tv\| \leq \|T\| \|v\|$$

Proposition 7.47. Suppose $T \in \mathcal{L}(V, W)$, then

- $\|T\| \geq 0$
- $\|T\| = 0 \iff T = 0$
- $\|\lambda T\| = |\lambda| \|T\|$ for all $\lambda \in \mathbf{F}$
- $\|S + T\| \leq \|S\| + \|T\|$ for all $S \in \mathcal{L}(V, W)$

Proposition 7.48. Suppose $T \in \mathcal{L}(V, W)$, then $\|T^*\| = \|T\|$.

Proposition 7.49. Suppose $T \in \mathcal{L}(V, W)$ and $s_1 \geq \cdots \geq s_m$ are the positive singular values of T. Suppose $1 \leq k \leq m$, then

$$\min \{\|T - S\| \mid S \in \mathcal{L}(V, W), \dim \text{range } S \leq k\} = s_{k+1}$$

Furthermore, if

$$Tv = s_1 \langle v, e_1 \rangle f_1 + \cdots + s_m \langle v, e_m \rangle f_m$$

is the SVD of T and $T_k \in \langle V, W \rangle$ is defined by

$$T_k v = s_1 \langle v, e_1 \rangle f_1 + \cdots + s_k \langle v, e_k \rangle f_k$$

for each $v \in V$, then $\dim \text{range } T_k = k$ and $\|T - T_k\| = s_{k+1}$

Proposition 7.50. Suppose $T \in \mathcal{L}(V)$. Then there exists a unitary operator $S \in \mathcal{L}(V)$ such that

$$T = S\sqrt{T^*T}$$

Remark. An anology of $z = \frac{z}{|z|}\sqrt{\bar{z}z}$ for every nonzero $z \in \mathbf{C}$.

Remark. If the SVD of T is

$$Tv = s_1\langle v, e_1\rangle f_1 + \cdots + s_m\langle v, e_m\rangle f_m$$

Then

$$T * Tv = s_1^2\langle v, e_1\rangle e_1 + \cdots + s_m^2\langle v, e_m\rangle e_m$$

and

$$\sqrt{T * T}v = s_1\langle v, e_1\rangle e_1 + \cdots + s_m\langle v, e_m\rangle e_m$$

Then let S be

$$Sv = \langle v, e_1\rangle f_1 + \cdots + \langle v, e_n\rangle f_n$$

can work.

Definition 7.51. Ball
The ball in V of radius 1 centered at 0 denoted by B, is defined by

$$B = \{v \in V \mid \|v\| < 1\}$$

Definition 7.52. Ellipsoid & Principal axes
Suppose that $f_1, \ldots, f_n$ is an orthonormal basis of V and $s_1, \ldots, s_n \in \mathbf{R}^+$. Then the ellipsoid with principle axes $s_1f_1, \ldots, s_nf_n$, denoted by $E(s_1f_1, \ldots, s_nf_n)$, is defined by

$$E(s_1f_1, \ldots, s_nf_n) = \left\{ v \in V \;\middle|\; \frac{|\langle v, f_1\rangle|^2}{s_1^2} + \cdots + \frac{|\langle v, f_n\rangle|^2}{s_n^2} < 1 \right\}$$

Remark. By the second point of Proposition 6.19, we have

$$E(f_1, \ldots, f_n) = B$$

Notation 7.53. $T(\Omega)$
For T a function defined on V and $\Omega \subset V$, $T(\Omega)$ is defined by

$$T(\Omega) = \{Tv \mid v \in \Omega\}$$

Remark. $T(V) = \text{range } T$

Proposition 7.54. Suppose $T \in \mathcal{L}(V)$ is invertible, then $T(B)$ is an ellipsoid in V.

Remark. Consider the SVD of T.

Remark. Invertible $\implies s_i > 0$

Proposition 7.55. Suppose $T \in \mathcal{L}(V)$ is invertible and E is an ellipsoid in V. Then $T(E)$ is an ellipsoid in V.

Remark. Consider $E = S(B)$ for an invertible $S \in \mathcal{L}(V)$.

Notation 7.56. $P(v_1, \ldots, v_n)$
Suppose $v_1, \ldots, v_n$ is a basis of V. Then

$$P(v_1, \ldots, v_n) = \{a_1 v_1 + \cdots + a_n v_n \mid a_1, \ldots, a_n \in (0, 1)\}$$

Definition 7.57. Parallelepiped & Edge
A parallelepiped is a set of the form $u + P(v_1, \ldots, v_n)$ for some $u \in V$, where $v_1, \ldots, v_n$ are called the edges of this parallelepiped.

Proposition 7.58. Suppose $u \in V$ and $v_1, \ldots, v_n$ is a basis of V. Suppose $T \in \mathcal{L}(V)$ is invertible. Then

$$T(u + P(v_1, \ldots, v_n)) = Tu + P(Tv_1, \ldots, Tv_n)$$

is also a parallelepiped.

Definition 7.59. Box
A box in V is a set of the form

$$u + P(r_1 e_1 + \cdots + r_n e_n)$$

where $u \in V$ and $r_1, \ldots, r_n$ are positive numbers and $e_1, \ldots, e_n$ is an orthonormal basis of V.

Proposition 7.60. Suppose $T \in \mathcal{L}(V)$ is invertible and has the singular value decomposition

$$Tv = s_1 \langle v, e_1 \rangle f_1 + \cdots + s_m \langle v, e_m \rangle f_m$$

Then

$$T(u + P(r_1 e_1 + \cdots + r_n e_n)) = Tu + P(r_1 s_1 f_1, \ldots, r_n s_n f_n)$$

is also a box.

Definition 7.61. Volume of a box
Suppose $\mathbf{F} = \mathbf{R}$. If $u \in V$ and $r_1, \ldots, r_n$ are positive numbers and $e_1, \ldots, e_n$ is an orthonormal basis of V, then

$$\text{volume}\,(u + P(r_1 e_1 + \cdots + r_n e_n)) = r_1 \cdots r_n$$

Definition 7.62. Volume
Suppose $\mathbf{F} = \mathbf{R}$. Then the volume of Ω, denoted by volume Ω, is approximately the sum of the volumes of a collection of disjoint boxes that approximate Ω.

Proposition 7.63. Suppose $\mathbf{F} = \mathbf{R}$, $T \in \mathcal{L}(V)$ is invertible, $\Omega \subset V$. Then

$$\frac{\text{volume } T(\Omega)}{\text{volume } \Omega} = \text{product of singular values of } T$$

Proposition 7.64. Suppose $T \in \mathcal{L}(V)$ is a normal operator with eigenvalues $\lambda_1, \ldots, \lambda_m$. Then

- T is invertible $\iff$ $\lambda_i \neq 0$
- T is self-adjoint $\iff$ $\lambda_i \in \mathbf{R}$
- T is positive $\iff$ $\lambda_i \in [0, \infty)$
- T is unitary $\iff$ $|\lambda_i| = 1$
- $\|T\| < 1 \iff |\lambda_i| < 1$

8 Operators on Complex Vector Spaces

8.1 Generalized Eigenvectors and Nilpotent Operators

Proposition 8.1. Suppose $T \in \mathcal{L}(V)$, then

$$\{0\} = \text{null } T^0 \subset \text{null } T^1 \subset \text{null } T^2 \subset \cdots$$

Proposition 8.2. Suppose $T \in \mathcal{L}(V)$ and $m \in \mathbf{N}$ such that

$$\text{null } T^m = \text{null } T^{m+1}$$

Then

$$\text{null } T^m = \text{null } T^{m+k}$$

for all $k \in \mathbf{N}$.

Remark.

$T^{m+2}v = 0 \implies Tv \in \text{null } T^{m+1} = \text{null } T^m \implies T^{m+1}v = 0 \implies \text{null } T^{m+2} = \text{null } T^{m+1}$

thus proved inductively.

Proposition 8.3. Suppose $T \in \mathcal{L}(V)$, then

$$\text{null } T^{\dim V} = \text{null } T^{\dim V+1} = \text{null } T^{\dim V+2} = \cdots$$

Remark.

$$\text{null } T^m \subsetneq \text{null } T^{m+1} \implies \dim \text{null } T^{m+1} > \dim \text{null } T^m$$

Proposition 8.4. Suppose $T \in \mathcal{L}(V)$, then

$$V = \text{null } T^{\dim V} \oplus \text{range } T^{\dim V}$$

Definition 8.5. Generalized eigenvector
Suppose $T \in \mathcal{L}(V)$ and λ is an eigenvalue of T. Then $v \in V$ is called a generalized eigenvector of T corresponding to λ iff $v \neq 0$ and

$$(T - \lambda I)^{\dim V} v = 0$$

Proposition 8.6. Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$, then there is a basis of V consisting of generalized eigenvectors of T.

Proposition 8.7. Suppose $T \in \mathcal{L}(V)$. Then each generalized eigenvector of T correspond to only one eigenvalue of T.

Remark. Consider the binomial expansion.

Proposition 8.8. Suppose that $T \in \mathcal{L}(V)$, then every list of generalized eigenvectors of T corresponding to distinct eigenvalues of T is linearly independent.

Remark. Referring to Proposition 5.6.

Definition 8.9. Nilpotent
An operator is called nilpotent if some power of it equals 0.

Remark. Suppose $T \in \mathcal{L}(V)$, then T is nilpotent iff $T^{\dim V} = 0$

Proposition 8.10. Suppose $T \in \mathcal{L}(V)$, then

- T is nilpotent $\implies$ 0 is the only eigenvalue of T.
- $\mathbf{F} = \mathbf{C}$ and 0 is the only eigenvalue of $T \implies T$ is nilpotent.

Proposition 8.11. Suppose $T \in \mathcal{L}(V)$, then the following are equivalent.

- T is nilpotent.
- The minimal polynomial of T is z^m for some positive integer m.
- There is a basis of V with respect to which the matrix of T has the form

$$\begin{pmatrix} 0 & & * \\ & \ddots & \\ 0 & & 0 \end{pmatrix}$$

  where all entries on and below the diagonal equal 0.

8.2 Generalized Eigenspace Decomposition

Definition 8.12. Generalized eigenspace
Suppose $T \in \mathcal{L}(V)$. The generalized eigenspace of T corresponding to λ, denoted by $G(\lambda, T)$, is defined by

$$G(\lambda, T) = \operatorname{null}(T - \lambda I)^{\dim V}$$

In other words, $G(\lambda, T)$ is the set of generalized eigenvectors of T corresponding to λ, along with the 0 vector.

Proposition 8.13. Generalized Eigenspace Decomposition
Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of T, then

- $G(\lambda_k, T)$ is invariant under T, for all $k \in \{1, \ldots, m\}$
- $(T - \lambda_k I)|_{G(\lambda_k, T)}$ is nilpotent for all $k \in \{1, \ldots, m\}$
- $V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$

Proposition 8.14. Suppose $T \in \mathcal{L}(V)$ and $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues of T, then

$$V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$$

iff the minimal polynomial of T equals

$$(z - \lambda_1)^{k_1} \cdots (z - \lambda_m)^{k_m}$$

for some $k_1, \ldots, k_m \in \mathbf{Z}^+$

Remark. $G(\lambda_1, T) + \cdots + G(\lambda_m, T)$ is a direct sum simply from Proposition 8.8.

Definition 8.15. Multiplicity
Suppose $T \in \mathcal{L}(V)$. Then the multiplicity of an eigenvalue λ of T equals

$$\dim G(\lambda, T) = \dim \operatorname{null}(T - \lambda I)^{\dim V}$$

Proposition 8.16. Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Then the sum of multiplicity of all eigenvalues of T equals $\dim V$.

Definition 8.17. Characteristic polynomial on C
Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Let $\lambda_1, \ldots, \lambda_m$ denote the distinct eigenvalues of T, with multiplicities $d_1, \ldots, d_m$. Then the characteristic polynomial of T is a polynomial

$$(z - \lambda_1)^{d_1} \cdots (z - \lambda_m)^{d_m}$$

Theorem 8.18. Cayley-Hamilton Theorem
Suppose $\mathbf{F} = \mathbf{C}$, $T \in \mathcal{L}(V)$, and q is the characteristic polynomial of T, then $q(T) = 0$.

Remark. This is equivalent to say the characteristic polynomial is a multiple of the minimal polynomial.

Proposition 8.19. Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Suppose $v_1, \ldots, v_n$ is a basis of V such that $\mathcal{M}(T, (v_1, \ldots, v_n))$ is upper triangular. Then the number of times that each eigenvalue of λ of T appears on the diagonal of $\mathcal{M}(T, (v_1, \ldots, v_n))$ equals the multiplicity of λ as an eigenvalue of T.

Definition 8.20. Block diagonal matrix
A block diagonal matrix is a square matrix of the form

$$\begin{pmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_m \end{pmatrix}$$

where $A_1, \ldots, A_m$ are square matrices lying along the diagonal and all other entries of the matrix equal 0.

Example 8.21. For example, a block diagonal matrix can be

$$\begin{pmatrix} \begin{pmatrix} 4 \end{pmatrix} & \begin{matrix} 0 & 0 \end{matrix} \\ \begin{matrix} 0 \\ 0 \end{matrix} & \begin{pmatrix} 6 & 3 \\ -1 & 2 \end{pmatrix} \end{pmatrix}$$

Proposition 8.22. Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of T, with multiplicities $d_1, \ldots, d_m$. Then there is a basis of V with respect to which T has a block matrix of the form

$$\begin{pmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_m \end{pmatrix}$$

where each A_k is a d_k-by-d_k upper-triangular matrix of the form

$$A_k = \begin{pmatrix} \lambda_k & & * \\ & \ddots & \\ 0 & & \lambda_k \end{pmatrix}$$

Remark. With generalized eigenspace decomposition, we can first address the case of each specific generalized eigenspace, or the case of a nilpotent, as shown in the second point in Proposition 8.13, and then extend it to all operators in the complex vector space. We will see some applications in the next subsection.

8.3 Consequences of Generalized Eigenspace Decomposition

Proposition 8.23. Suppose $T \in \mathcal{L}(V)$ is nilpotent. Then $I + T$ has a square root.

Remark. Consider the first $\dim V$ terms of the talor expansion of $\sqrt{1+x}$ gives

$$I + \frac{1}{2}T - \frac{1}{8}T^2 + \cdots + aT^{\dim V - 1}$$

is a square root of $I + T$ since its square equals

$$I + T + T^{\dim V}(\cdots) = I + T$$

Proposition 8.24. Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is invertible. Then T has a square root.

Definition 8.25. Jordan basis
Suppose $T \in \mathcal{L}(V)$. A basis of V is called a Jordan basis for T if with respect to this basis T has a block diagonal matrix

$$\begin{pmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_p \end{pmatrix}$$

in which each A_k is an upper-triangular matrix of the form

$$\begin{pmatrix} \lambda_k & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda_k \end{pmatrix}$$

Remark. Note that λ_k need not be distinct.

Proposition 8.26. Suppose $T \in \mathcal{L}(V)$ is nilpotent. Then there is a basis of V that is a Jordan basis for T.

Theorem 8.27. Jordan Form
Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Then there is a basis of V that is a Jordan basis for T.

8.4 Trace: A Connection Between Matrices and Operators

Definition 8.28. Trace of a matrix
Suppose A is a square matrix with entries in $\mathbf{F}$. The trace of A, denoted by $\operatorname{tr} A$, is defined to be the sum of the diagonal entries of A.

Proposition 8.29. Suppose A is an n-by-m matrix and B is an m-by-n matrix. Then

$$\operatorname{tr}(AB) = \operatorname{tr}(BA)$$

Proposition 8.30. Suppose $T \in \mathcal{L}(V)$. Suppose also $v_1, \ldots, v_n$ and $u_1, \ldots, u_n$ are bases of V, then

$$\operatorname{tr} \mathcal{M}(T, (v_1, \ldots, v_n)) = \operatorname{tr} \mathcal{M}(T, (u_1, \ldots, u_n))$$

Remark. This shows that the trace of a matrix of an operator does not depend on basis.

Definition 8.31. Trace of an operator
Suppose $T \in \mathcal{L}(V)$. Then the trace of T, denoted $\operatorname{tr} T$, is defined by

$$\operatorname{tr} T = \operatorname{tr} \mathcal{M}(T, (v_1, \ldots, v_n))$$

where $v_1, \ldots, v_n$ is any basis of V.

Remark. Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$, suppose also that $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of T, with multiplicities $d_1, \ldots, d_m$. Then we can intentionally take a basis of V such that $\mathcal{M}(V)$ is a upper-triangular matrix, and then by Proposition 8.19, we have

$$\operatorname{tr} T = d_1\lambda_1 + \cdots + d_m\lambda_m$$

Proposition 8.32. Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Then $\operatorname{tr} T$ equals the negative of the coefficient of $z^{\dim V - 1}$ in the characteristic polynomial of T.

Proposition 8.33. Suppose $T \in \mathcal{L}(V)$ and $e_1, \ldots, e_n$ is an orthonormal basis of V. Then

$$\operatorname{tr} T = \langle Te_1, e_1 \rangle + \cdots + \langle Te_n, e_n \rangle$$

Remark. Notice that $\mathcal{M}(T, (e_1, \ldots, e_n))_{k,k} = \langle Te_k, e_k \rangle$ by the first point of Proposition 6.19.

Proposition 8.34. The function $\operatorname{tr} : \mathcal{L}(V) \to \mathbf{F}$ is a linear functional on $\mathcal{L}(V)$.

Corollary 8.35. There do not exist $S, T \in \mathcal{L}(V)$ such that $ST - TS = I$.

Remark. Note that $\operatorname{tr} I = \dim V$.

9 Multilinear Algebra and Determinants

9.1 Bilinear Forms and Quadratic Forms

Definition 9.1. Bilinear Form
A bilinear form on V is a function $\beta : V \times V \to \mathbf{F}$ such that

$$v \mapsto \beta(v, u) \quad \text{and} \quad v \mapsto \beta(u, v)$$

are both linear functionals on V for all fixed $u \in V$.

Remark. $\langle v, v \rangle : V \times V \to \mathbf{F}$ is a bilinear form if $\mathbf{F} = \mathbf{R}$, but fails when $\mathbf{F} = \mathbf{C}$.

Notation 9.2. $V^{(2)}$
The set of bilinear forms on V is denoted by $V^{(2)}$.

Definition 9.3. Matrix of a bilinear form
Suppose $\beta \in V^{(2)}$ and $v_1, \ldots, v_n$ is a basis of V. Then the matrix of β with respect to this basis is the n-by-n matrix $\mathcal{M}(\beta)$ defined by

$$\mathcal{M}(\beta)_{i,j} = \beta(v_i, v_j)$$

Proposition 9.4. Suppose $\beta \in V^{(2)}$, then $\beta \mapsto \mathcal{M}(\beta)$ is an isomorphism of $V^{(2)}$ onto $\mathbf{F}^{n,n}$.

Remark. As a result, we have

$$\dim V^{(2)} = (\dim V)^2$$

Proposition 9.5. Suppose $\beta \in V^{(2)}$ and $T \in \mathcal{L}(V)$. Defined $\alpha, \rho \in V^{(2)}$ by

$$\alpha(u, v) = \beta(u, Tv) \quad \text{and} \quad \rho(u, v) = \beta(Tu, v)$$

Then

$$\mathcal{M}(\alpha) = \mathcal{M}(\beta)\mathcal{M}(T) \quad \text{and} \quad \mathcal{M}(\rho) = \mathcal{M}(T)^{\mathrm{t}}\mathcal{M}(\beta)$$

Proposition 9.6. Suppose $\beta \in V^{(2)}$. Suppose also that $e_1, \ldots, e_n$ and $f_1, \ldots, f_n$ are bases of V. Let

$$A = \mathcal{M}(\beta, (e_1, \ldots, e_n)) \quad \text{and} \quad B = \mathcal{M}(\beta, (f_1, \ldots, f_n))$$

Suppose $C = \mathcal{M}(I, (e_1, \ldots, e_n), (f_1, \ldots, f_n))$, then

$$A = C^{\mathrm{t}} B C$$

Definition 9.7. Symmetric bilinear form
A bilinear form $\rho \in V^{(2)}$ is called symmetric if

$$\rho(u, w) = \rho(w, u)$$

for all $u, v \in V$.

Notation 9.8. $V^{(2)}_{\mathrm{sym}}$
The set of all symmetric bilinear form on V is denoted $V^{(2)}_{\mathrm{sym}}$.

Definition 9.9. Symmetric matrix
A sqaure matrix A is called symmetric if $A = A^{\mathrm{t}}$.

Proposition 9.10. Suppose $\rho \in V^{(2)}$. Then the following are equivalent.

- ρ is symmetric.
- $\mathcal{M}(\rho, (e_1, \ldots, e_n))$ is a symmetric matrix for every basis $e_1, \ldots, e_n$ of V.
- $\mathcal{M}(\rho, (e_1, \ldots, e_n))$ is a symmetric matrix for some basis $e_1, \ldots, e_n$ of V.
- $\mathcal{M}(\rho, (e_1, \ldots, e_n))$ is a diagonal matrix for some basis $e_1, \ldots, e_n$ of V.

Proposition 9.11. Suppose V is a real inner product space and ρ is a symmetric biliear form on V. Then ρ has a diagonal matrix with respect to some orthonormal basis of V.

Definition 9.12. Alternating bilinear form
A bilinear form $\alpha \in V^{(2)}$ is called alternating if

$$\alpha(v, v) = 0$$

for all $v \in \mathcal{L}(V)$.

Notation 9.13. $V^{(2)}_{\text{alt}}$
The set of alternating bilinear forms on V is denoted by $V^{(2)}_{\text{alt}}$.

Proposition 9.14. A bilinear form α on V is alternating iff

$$\alpha(u, w) = -\alpha(w, u)$$

for all $u, w \in V$.

Remark. Consider $0 = \alpha(u + w, u + w) = \alpha(u, w) + \alpha(w, u)$.

Proposition 9.15. $V^{(2)}_{\text{sym}}$ and $V^{(2)}_{\text{alt}}$ are subspaces of $V^{(2)}$. Furthermore,

$$V^{(2)} = V^{(2)}_{\text{sym}} \oplus V^{(2)}_{\text{alt}}$$

Remark. Consider

$$\alpha(u, w) = \frac{\alpha(u, w) + \alpha(w, u)}{2} + \frac{\alpha(u, w) - \alpha(w, u)}{2}$$

where

$$\frac{\alpha(u, w) + \alpha(w, u)}{2} \in V^{(2)}_{\text{sym}} \quad \text{and} \quad \frac{\alpha(u, w) - \alpha(w, u)}{2} \in V^{(2)}_{\text{alt}}$$

Remark. The above definitions and results are rather analogous to even functions and odd functions:

- Even functions defined by $f(x) = f(-x)$ compared with symmetric bilinear form defined by $\rho(u, w) = \rho(w, u)$
- Odd functions defined by $f(x) = -f(-x)$ compared with alternating bilinear form where $\alpha(u, w) = -\alpha(w, u)$ holds.
- Similarly, every function can be written as the sum of an odd function and an even function by considering

$$f(x) = \frac{f(x) + f(-x)}{2} + \frac{f(x) - f(-x)}{2}$$

Definition 9.16. Quadratic form
For a bilinear form $\beta \in V^{(2)}$, the quadratic form associated with it, denoted $q_\beta : V \to \mathbf{F}$, is defined by

$$q_\beta(v) = \beta(v, v)$$

for all $v \in V$.
A function $q : V \to \mathbf{F}$ is a quadratic form on V if there exist $\beta \in V^{(2)}$ such that $q = q_\beta$.

Proposition 9.17. Suppose $n \in \mathbf{Z}^+$ and $q : \mathbf{F}^n \to \mathbf{F}$. Then q is a quadratic form on $\mathbf{F}^n$ iff there exist $A_{i,j} \in \mathbf{F}$ for all $i, j \in \{1, \ldots, n\}$ such that

$$q(x_1, \ldots, x_n) = \sum_{j=1}^{n} \sum_{i=1}^{n} A_{i,j} x_i x_j$$

for all $(x_1, \ldots, x_n) \in \mathbf{F}^n$.

Proposition 9.18. Suppose $q : V \to \mathbf{F}$ is a function. Then the following are equivalent.

- q is a quadratic form.
- There exists a unique symmetric bilinear form ρ on V such that $q = q_\rho$.
- $q(\lambda v) = \lambda^2 q(v)$ for all $\lambda \in \mathbf{F}$ and all $v \in V$, and the function

  $$(u, w) \mapsto q(u + w) - q(u) - q(w)$$

  is a symmetric bilinear form on V.
- $q(2v) = 4q(v)$ for all $v \in V$, and the function

  $$(u, w) \mapsto q(u + w) - q(u) - q(w)$$

  is a symmetric bilinear form on V.

Remark. The second point reveals that for a quadratic form $q = q_\beta$, we have

$$\{\alpha \mid q = q_\alpha\} = \left\{ \beta + \theta \,\middle|\, \theta \in V^{(2)}_{\text{alt}} \right\}$$

Proposition 9.19. Suppose q is a quadratic form on V, then there exist a basis $e_1, \ldots, e_n$ and $\lambda_1, \ldots, \lambda_n \in \mathbf{F}$ such that

$$q(x_1e_1 + \ldots, x_ne_n) = \lambda_1 x_1^2 + \cdots + \lambda_n x_n^2$$

for all $x_1, \ldots, x_n \in \mathbf{F}$.
Furthermore, $e_1, \ldots, e_n$ can be an orthonormal basis if $\mathbf{F} = \mathbf{R}$.

Remark. We can consider the symmetric bilinear form ρ of q. Then the proposition is directly proved by the last point of Proposition 9.10 and Proposition 9.11.

9.2 Alternating Multilinear Forms

Definition 9.20. *m-linear form*
For $m \in \mathbf{Z}^+$, a m-linear form on V is a function $\beta : V^m \to \mathbf{F}$ that is linear in each slot when other slots are held fixed. In other words, for all $k \in \{1, \ldots, m\}$ and all $u_1, \ldots, u_m \in V$. The function

$$v \mapsto \beta(u_1, \ldots, u_{k-1}, v, u_{k+1}, \ldots, u_m)$$

is a linear functional from V to $\mathbf{F}$.

Notation 9.21. $V^{(m)}$
The set of m-linear forms on V is denoted by $V^{(m)}$.

Definition 9.22. Multilinear form
A function β is a multilinear form if it is an m-linear form on V for some $m \in \mathbf{Z}^+$.

Definition 9.23. Alternating multilinear forms
Suppose $m \in \mathbf{Z}^+$, then an m-linear form α on V is called alternating if

$$\exists i \neq j (v_i = v_j) \implies \alpha(v_1, \ldots, v_m) = 0$$

Notation 9.24. $V_{\text{alt}}^{(m)}$
The set of alternating m-linear forms on V is denoted by $V_{\text{alt}}^{(m)}$.

Proposition 9.25. Suppose $m \in \mathbf{Z}^+$ and $\alpha \in V_{\text{alt}}^{(m)}$. If $v_1, \ldots, v_m$ is linearly dependent, then

$$\alpha(v_1, \ldots, v_m) = 0$$

Corollary 9.26. Suppose $m > \dim V$. Then $V_{\text{alt}}^{(m)} = \{0\}$.

Definition 9.27. Permutation
Suppose $m \in \mathbf{Z}^+$, then a permutation of $(1, \ldots, m)$ is a list $(a_1, \ldots, a_m)$ that contains each of the numbers $1, \ldots, m$ exactly once.

Notation 9.28. $\operatorname{perm} m$
The set of all permutations of $(1, \ldots, m)$ is denoted by $\operatorname{perm} m$

Definition 9.29. Sign of a permutation
The sign of a permutation $(a_1, \ldots, a_m)$ is defined by

$$\operatorname{sign}(a_1, \ldots, a_m) = (-1)^N$$

where N is the number of pairs of integers (k, l) with $1 \leq k < l \leq m$ such that k appears after l in the list $(a_1, \ldots, a_m)$.

Proposition 9.30. Swapping two entries in a permutation multiplies the sign of the permutation by -1.

Proposition 9.31. Suppose $m \in \mathbf{Z}^+$ and $\alpha \in V_{\text{alt}}^{(m)}$, then

$$\alpha(v_{a_1}, \ldots, v_{a_m}) = \operatorname{sign}(a_1, \ldots, a_m)\alpha(v_1, \ldots, v_m)$$

for all list $v_1, \ldots, v_m$ of vectors in V and all $(a_1, \ldots, a_m) \in \operatorname{perm} m$.

Proposition 9.32. Suppose $\dim V = n$, $\alpha \in V_{\text{alt}}^{(n)}$, $e_1, \ldots, e_n$ is a basis of V and $v_1, \ldots, v_n \in V$. For each $k \in \{1, \ldots, n\}$, let $b_{1,k}, \ldots, b_{n,k} \in \mathbf{F}$ be such that

$$v_k = \sum_{i=1}^{n} b_{i,k} e_i$$

Then we have

$$\alpha(v_1, \ldots, v_n) = \alpha(e_1, \ldots, e_n) \cdot \left(\sum_{(i_1, \ldots, i_n) \in \operatorname{perm} n} \operatorname{sign}(i_1, \ldots, i_n) b_{i_1,1} \cdots b_{i_n,n} \right)$$

Remark. Consider the expansion of

$$(b_{1,1}e_1 + \cdots + b_{n,1}e_n) \cdots (b_{1,n}e_1 + \cdots b_{n,n}e_n)$$

where the terms with a power of e_i greater than 1 is removed (by Definition 9.23).

Proposition 9.33.

$$\dim V_{\text{alt}}^{(\dim V)} = 1$$

Remark. We can show from Proposition 9.32 that $\alpha' = c\alpha$ for any $\alpha, \alpha' \in V_{\text{alt}}^{(\dim V)}$.

Proposition 9.34. Suppose $\dim V = n$, α is a nonzero alternating n-linear form on V and $e_1, \ldots, e_n$ is a list of vectors in V. Then

$$\alpha(e_1, \ldots, e_n) \neq 0 \iff e_1, \ldots, e_n \text{ is linearly independent}$$

Remark. Note that $e_1, \ldots, e_n$ is linear independent is equivalent to $e_1, \ldots, e_n$ is a basis of V.

9.3 Determinants

Definition 9.35. α_T
Suppose $m \in \mathbf{Z}^+$ and $T \in \mathcal{L}(V)$. For $\alpha \in V_{\text{alt}}^{(m)}$, define $\alpha_T \in V_{\text{alt}}^{(m)}$ by

$$\alpha_T(v_1, \ldots, v_m) = \alpha(Tv_1, \ldots, Tv_m)$$

for each list $v_1, \ldots, v_n$ of vectors in V.

Definition 9.36. Determinant of an operator
Suppose $T \in \mathcal{L}(V)$. Then the determinant of T, denoted by $\det T$, is defined by

$$\det T = \frac{\alpha_T}{\alpha}$$

for all $\alpha \in V_{\text{alt}}^{(\dim V)}$.

Remark. To prove that $\det T$ is well-defined, i.e. the uniqueness of $\frac{\alpha_T}{\alpha}$, we notice that as expressed in Proposition 9.32, $\frac{\alpha'}{\alpha}$ is independent of α, only the relationship between variables matters.
In other words, if we have $Te_i = \sum_{j=1}^{n} A_{i,j}e_j$ for each $i \in \{1, \ldots, n\}$, then

$$\det T = \frac{\alpha_T}{\alpha} = \sum_{(j_1,\ldots,j_n)\in \text{perm}\, n} \text{sign}(j_1, \ldots, j_n) A_{j_1,1} \cdots A_{j_n,n}$$

Example 9.37. Let $n = \dim V$, then

- $\det I = 1$
- $\det(\lambda T) = \lambda^n \det T$
- Suppose $T \in \mathcal{L}(V)$ and there is a basis $e_1, \ldots, e_n$ of V consisting of eigenvectors of T, with corresponding eigenvalues $\lambda_1, \ldots, \lambda_n$. Then

$$\det T = \lambda_1 \cdots \lambda_n$$

Definition 9.38. Determinant of a matrix
Suppose that A is a n-by-n square matrix with entries in $\mathbf{F}$. Let $T \in \mathcal{L}(\mathbf{F}^n)$ be the operator whose matrix with respect to the standard basis of $\mathbf{F}^n$ equals A. Then the determinant of A, denoted by $\det A$, is defined by $\det A = \det T$.

Remark. Note that we now have $Te_i = \sum_{j=1}^{n} A_{i,j} e_j$ for each $i \in \{1, \ldots, n\}$, where $e_1, \ldots, e_n$ is the standard basis. Hence by Remark in Definition 9.36, we have

$$\det A = \det T = \sum_{(j_1,\ldots,j_n)\in \operatorname{perm} n} \operatorname{sign}(j_1, \ldots, j_n) A_{j_1,1} \cdots A_{j_n,n}$$

Proposition 9.39. Suppose $v_1, \ldots, v_n \in \mathbf{F}^n$, then the function $\alpha : (\mathbf{F}^n)^n \to \mathbf{F}$

$$\alpha(v_1, \ldots, v_n) = \det \begin{pmatrix} v_1 & \ldots & v_n \end{pmatrix}$$

is an alternating n-linear form on $\mathbf{F}^n$.

Remark. The operator correspond to the matrix $\begin{pmatrix} v_1 & \ldots & v_n \end{pmatrix}$ is $Te_k = v_k$.

Example 9.40. Suppose that A is an upper-triangular matrix with $\lambda_1, \ldots, \lambda_n$ on the diagonal, then

$$\det A = \lambda_1 \cdots \lambda_n$$

Proposition 9.41. Suppose $S, T \in \mathcal{L}(V)$. Then $\det(ST) = (\det S)(\det T)$.

Remark. This shows that determinant is multiplicative.

Corollary 9.42. An operator $T \in \mathcal{L}(V)$ is invertible iff $\det T \neq 0$. Furthermore, if T is invertible, then $\det T^{-1} = \frac{1}{\det T}$.

Proposition 9.43. Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbf{F}$. Then λ is an eigenvalue of T iff $\det(\lambda I - T) = 0$.

Proposition 9.44. Suppose $T \in \mathcal{L}(V)$ and $S : W \to V$ is an invertible linear map. Then

$$\det(S^{-1}TS) = \det(T)$$

Proposition 9.45. Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Suppose also that the eigenvalues of T is $\lambda_1, \ldots, \lambda_m$, with multiplicities $d_1, \ldots, d_m$. Then

$$\det T = \lambda_1^{d_1} \cdots \lambda_m^{d_m}$$

Proposition 9.46.

- Suppose A is a square matrix. Then $\det A^{\mathrm{t}} = \det A$.
- Suppose $T \in \mathcal{L}(V)$. Then $\det T' = \det T$.
- Suppose $T \in \mathcal{L}(V)$. Then $\det(T^*) = \overline{\det T}$.

Proposition 9.47.

- If either two columns or two rows of a square matrix are equal, then the determinant of the matrix equals 0.
- Suppose A is a square matrix and B is the matrix obtained from A by swapping either two columns or two rows. Then $\det A = -\det B$.
- If one column or one row of a square matrix is multiplied by a scalar, then the value of the determinant is multiplied by the same scalar.

- If a scalar multiple of one column of a square matrix to added to another column, then the value of the determinant is unchanged.
- If a scalar multiple of one row of a square matrix to added to another row, then the value of the determinant is unchanged.

Proposition 9.48. Suppose $S \in \mathcal{L}(V)$ is a unitary operator. Then $|\det S| = 1$.

Remark. Note that $SS^* = I$ as proved in Proposition 7.30.

Proposition 9.49. Suppose $T \in \mathcal{L}(V)$ is a positive operator. Then $\det T \geq 0$.

Remark. Note that all eigenvalues of T is nonnegative by Proposition 7.24 and that T is diagonalizable by Spectral theorem.

Proposition 9.50. Suppose $T \in \mathcal{L}(V)$. Then

$$|\det T| = \text{product of singular values of } T$$

Remark. Note that $\det T = \sqrt{\det T^*T}$ and that the eigenvalues of T^*T are $s_1^2, \ldots, s_n^2$.

Proposition 9.51. Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Suppose also that $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of T, with multiplicities $d_1, \ldots, d_m$. Then

$$\det(zI - T) = (z - \lambda_1)^{d_1} \cdots (z - \lambda_m)^{d_m}$$

Definition 9.52. Characteristic polynomial
Suppose $T \in \mathcal{L}(V)$. Then the characteristic polynomial is defined by

$$p(z) = \det(zI - T)$$

Theorem 9.53. Cayley-Hamilton Theorem
Suppose $T \in \mathcal{L}(V)$ and q is the characteristic polynomial of T, then $q(T) = 0$.

Proposition 9.54. Suppose $T \in \mathcal{L}(V)$ and $\dim V = n$. Then the characteristic polynomial of T can be written as

$$z^n - (\operatorname{tr} T)z^{n-1} + \cdots + (-1)^n(\det T)$$

Proposition 9.55. Hadamard's inequality
Suppose A is an n-by-n matrix. Let $v_1, \ldots, v_n$ denote the columns of A. Then

$$|\det A| \leq \prod_{k=1}^{n} \|v_k\|$$

Remark. The equality holds iff the columns of A are pairwise orthogonal.

Proposition 9.56. Suppose $n > 1$ and $\beta_1, \ldots, \beta_n \in \mathbf{F}$. Then

$$\det \begin{pmatrix} 1 & \beta_1 & \beta_1^2 & \cdots & \beta_1^{n-1} \\ 1 & \beta_2 & \beta_2^2 & \cdots & \beta_2^{n-1} \\ & & \vdots & & \\ 1 & \beta_n & \beta_n^2 & \cdots & \beta_n^{n-1} \end{pmatrix} = \prod_{1 \leq i < j \leq n} (\beta_j - \beta_i)$$

Remark. The matrix in this proposition is called **Vandermonde matrix**.

9.4 Tensor Products

Definition 9.57. Bilinear functional
A bilinear functional on $V \times W$ is a function $\beta : V \times W \to \mathbf{F}$ such that $v \mapsto \beta(v, w)$ is a linear functional on V for each $w \in W$ and that $w \mapsto \beta(v, w)$ is also a linear functional on W for each $v \in V$.

Remark. Note that if $V = W$, then β is a linear form.

Notation 9.58. $\mathcal{B}(V, W)$
The vector space of bilinear functionals on $V \times W$ is denoted by $\mathcal{B}(V, W)$.

Proposition 9.59.

$$\dim \mathcal{B}(V, W) = (\dim V)(\dim W)$$

Remark. $\mathcal{B}(V, W)$ can be regarded as $\mathcal{L}(V, \mathcal{L}(W, \mathbf{F}))$.

Definition 9.60. Tensor product

- The tensor product $V \otimes W$ is defined to be $\mathcal{B}(V', W')$.
- For $v \in V$ and $w \in W$, the tensor product $v \otimes w \in V \otimes W$ is defined by

$$(v \otimes w)(\varphi, \tau) = \varphi(v)\tau(w)$$

for all $(\varphi, \tau) \in V' \times W'$.

Proposition 9.61.

$$\dim(V \otimes W) = (\dim V)(\dim W)$$

Proposition 9.62. Suppose $v, v_1, v_2 \in V$ and $w, w_1, w_2 \in W$ and $\lambda \in \mathbf{F}$. Then

$$(v_1 + v_2) \otimes w = v_1 \otimes w + v_2 \otimes w$$

and

$$v \otimes (w_1 + w_2) = v \otimes w_1 + v \otimes w_2$$

and

$$\lambda(v \otimes w) = (\lambda v) \otimes w = v \otimes (\lambda w)$$

Proposition 9.63. Suppose $e_1, \ldots, e_m$ and $f_1, \ldots, f_n$ are linearly independent lists of vectors in V and W respectively. Then

$$\{e_i \otimes f_j \mid i \in \{1, \ldots, m\}, j \in \{1, \ldots, n\}\}$$

is a linearly independent list in $V \otimes W$.
Furthermore, if $e_1, \ldots, e_m$ and $f_1, \ldots, f_n$ are bases of V and W, then

$$\{e_i \otimes f_j \mid i \in \{1, \ldots, m\}, j \in \{1, \ldots, n\}\}$$

is a basis of $V \otimes W$.

Remark. If $\{e_i \otimes f_j \mid i \in \{1, \ldots, m\}, j \in \{1, \ldots, n\}\}$ is linearly dependent, then there exists $A_{i,j} \in \mathbf{F}$ such that for all $\varphi \in V'$ and $\tau \in W'$,

$$\sum_{i=1}^{m} \sum_{j=1}^{n} A_{i,j}\varphi(e_i)\tau(f_j) = \sum_{i=1}^{m} (\sum_{j=1}^{n} A_{i,j}\tau(f_j))\varphi(e_i) = 0$$

which means either $\varphi(e_i)$ or $\tau(f_j)$ is a linearly dependent list.

Definition 9.64. Bilinear map
A bilinear map from $V \times W$ to a vector space U is a function $\Gamma : V \times W \to U$ such that $v \mapsto \Gamma(v, w)$ is a linear map from V to U for each $w \in W$ and $w \mapsto \Gamma(v, w)$ is a linear map from W to U for each $v \in V$.

Proposition 9.65. Suppose U is a vector space, then

- Suppose $\Gamma : V \times W \to U$ is a bilinear map. Then there exists a unique linear map $\hat{\Gamma} = V \otimes W \to U$ such that
$$\hat{\Gamma}(v \otimes w) = \Gamma(v, w)$$
for all $(v, w) \in V \times W$.
- Suppose $T : V \otimes W \to U$ is a linear map. Then there exists a unique bilinear map $T^{\#} : V \times W \to U$ such that
$$T^{\#}(v, w) = T(v \otimes w)$$
for all $(v, w) \in V \times W$.

Proposition 9.66. Suppose V and W are inner product spaces. Then there is a unique inner product on $V \otimes W$ such that
$$\langle v \otimes w, u \otimes x \rangle = \langle v, u \rangle \langle w, x \rangle$$
for all $v, u \in V$ and $w, x \in W$.

Definition 9.67. Inner product on tensor product
Suppose V and W are inner product spaces. The inner product on $V \otimes W$ is the unique function $\langle \cdot, \cdot \rangle : (V \otimes W) \times (V \otimes W) \to \mathbf{F}$ such that
$$\langle v \otimes w, u \otimes x \rangle = \langle v, u \rangle \langle w, x \rangle$$
for all $v, u \in V$ and $w, x \in W$.

Remark. Note if $e_1, \ldots, e_m$ is an orthonormal basis in V and $f_1, \ldots, f_n$ is an orthonormal basis in W, then the inner product is defined by
$$\langle \sum_{k=1}^{n} \sum_{j=1}^{m} b_{j,k} e_j \otimes f_k, \sum_{k=1}^{n} \sum_{j=1}^{m} c_{j,k} e_j \otimes f_k \rangle = \sum_{k=1}^{n} \sum_{j=1}^{m} b_{j,k} \overline{c_{j,k}}$$

Remark. Consider the last point in Proposition 6.19.

Remark. This shows that

$$\|v \otimes w\| = \|v\| \times \|w\|$$

Proposition 9.68. Suppose V and W are inner product spaces, and $e_1, \ldots, e_m$ is an orthonormal basis of V and $f_1, \ldots, f_n$ is an orthonormal basis of W, then

$$\{e_i \otimes f_j \mid i \in \{1, \ldots, m\}, j \in \{1, \ldots, n\}\}$$

is an orthonormal basis of $V \otimes W$.

Remark. The following definitions and propositions are purely generalization of the previous results.

Definition 9.69. *m-linear functional*
An m-linear functional on $V_1 \times \cdots \times V_m$ is a function $\beta : V_1 \times \cdots \times V_m \to \mathbf{F}$ that is a linear functional in each slot when the other slots are fixed.

Notation 9.70. $B(V_1, \ldots, V_m)$
We use $B(V_1, \ldots, V_m)$ to denote the vector space of m-linear functionals on $V_1 \times \cdots \times V_m$.

Proposition 9.71.

$$\dim \mathcal{B}(V_1, \ldots V_m) = (\dim V_1) \times \cdots \times (\dim V_m)$$

Definition 9.72. Generalized tensor product

- The tensor product $V_1 \otimes \cdots \otimes V_m$ is defined to be $\mathcal{B}(V_1', \cdots, V_m')$.
- For $v_1 \in V_1, \ldots, v_m \in V_m$, the tensor product $v_1 \otimes \cdots \otimes v_m \in V_1 \otimes \cdots \otimes V_m$ is defined by

  $$(v_1 \otimes \cdots \otimes v_m)(\varphi_1, \ldots, \varphi_m) = \varphi_1(v_1) \cdots \varphi_m(v_m)$$

  for all $(\varphi_1, \ldots, \varphi_m) \in V_1' \times \cdots \times V_m'$.

Proposition 9.73.

$$\dim(V_1 \otimes \cdots \otimes V_m) = (\dim V_1) \cdots (\dim V_m)$$

Proposition 9.74. Suppose $\dim V_k = n_k$ and $e_{k,1}, \cdots, e_{k,n_k}$ is a basis of V_k for each $k \in \{1, \ldots, m\}$, then

$$\{e_{1,j_1} \otimes \cdots \otimes e_{m,j_m} \mid j_i \in \{1, \ldots, n_i\}\}$$

is a basis of $V_1 \otimes \cdots \otimes V_m$.

Definition 9.75. ***m*-linear map**
An m-linear map from $V_1 \times \cdots \times V_m$ to a vector space U is a function $\Gamma : V_1 \times \cdots \times V_m \to U$ that is a linear map in each slot when the other slots are fixed.

Proposition 9.76. Suppose U is a vector space, then

- Suppose $\Gamma : V_1 \times \cdots \times V_m \to U$ is an m-linear map. Then there exists a unique linear map $\hat{\Gamma} : V_1 \otimes \cdots \otimes V_m \to U$ such that

  $$\hat{\Gamma}(v_1 \otimes \cdots \otimes v_m) = \Gamma(v_1, \ldots, v_m)$$

  for all $(v_1, \ldots, v_m) \in V_1 \times \cdots \times V_m$

- Suppose $T : V_1 \otimes \cdots \otimes V_m \to U$ is a linear map. Then there exists a unique m-linear map $T^{\#} : V_1 \times \cdots \times V_m \to U$ such that

  $$T^{\#}(v_1, \ldots, v_m) = T(v_1 \otimes \cdots \otimes v_m)$$

  for all $(v_1, \ldots, v_m) \in V_1 \times \cdots \times V_m$

Definition 9.77. Inner product on generalized tensor product
Suppose $V_1, \ldots, V_m$ are finite-dimensional inner product spaces. Then the inner product on $V_1 \otimes \cdots \otimes V_m$ is the unique function $\langle \cdot, \cdot \rangle : (V_1 \otimes \cdots \otimes V_m)^2 \to \mathbf{F}$ such that

$$\langle v_1 \otimes \cdots \otimes v_m, u_1 \otimes \cdots \otimes u_m \rangle = \langle v_1, u_1 \rangle \cdots \langle v_m, u_m \rangle$$

for all $(v_1, \ldots, v_m), (u_1, \ldots, u_m) \in V_1 \times \cdots \times V_m$

Corollary 9.78.

$$\|v_1 \otimes \cdots \otimes v_m\| = \|v_1\| \times \cdots \times \|v_m\|$$

Proposition 9.79. Suppose $V_1, \ldots, V_m$ are inner product spaces with $e_{i,1}, \ldots, e_{i,n_i}$ being orthonormal basis of V_i. Then

$$\{e_{1,j_1} \otimes \cdots \otimes e_{m,j_m} \mid j_i \in \{1, \ldots, n_i\}\}$$

is an orthonormal basis of $V_1 \otimes \cdots \otimes V_m$.

10 Glossary

Vector Space

$\mathbf{F}, V, W$

Vector Space

$\mathbf{F}^S$

Subspace

Sum of Subsets

Direct Sum

$U_1 \oplus U_2$

Finite-Dimensional Vector Spaces

Span

Spans

Finite & Infinite-dimensional vector space

$\mathcal{P}(\mathbf{F})$

$\deg p$

$\mathcal{P}_m(\mathbf{F})$

Linearly Independent & Dependent

Basis

Dimension

$\dim V$

Linear Maps

Linear map

$\mathcal{L}(V, W)$

Product of Linear Maps

Null Space

$\text{null } T$

Injective

Range

range T

Surjective

Matrix

Matrix of a linear map

$\mathcal{M}(T)$

Matrix Addition

Scalar multiplication of a matrix

$\mathbf{F}^{m,n}$

Matrix multiplication

$A_{j,.}$ & $A_{.,k}$

Invertible & Inverse

T^{-1}

Isomorphism & Isomorphic

Matrix of a vector

$\mathcal{M}(v)$

Operator

$\mathcal{L}(V)$

Product of vector spaces

$v + U$

Affine subset

Parallel

Quotient space

V/U

Quotient map

$\pi(v)$

$\tilde{T}$

Linear functional

Dual space

V'

Dual basis

Dual map

T'

Annihilator

U^0

Transpose

A^t

Row Rank & Column Rank

Rank

Polynomials

Zero of a polynomial

Factor

Eigenvalues and Eigenvectors

Invariant subspace

Eigenvalue

Eigenvector

Restriction operator & quotient operator

$T|_U$ & T/U

T^m

$p(T)$

Product of polynomials

Monic polynomial

Minimal polynomial

Matrix of an operator

$\mathcal{M}(T)$

Diagonal of a matrix

Upper-triangular matrix

Diagonal matrix

Eigenspace

$E(\lambda, T)$

Diagonalizable

Gershgorin disks

Commute

Inner Product Spaces

Dot Product

Inner Product

$\langle u, v \rangle$

Inner product space

Norm

$\|v\|$

Orthogonal

Orthonormal

Orthonormal basis

Orthogonal complement

$U^{\perp}$

Orthogonal projection

P_U

Pseudoinverse

$T^{\dagger}$

Operators on Inner Product Spaces

Adjoint

T^*

Conjugate transpose

A^*

Self-adjoint

Normal

Positive operator

Square root

$\sqrt{T}$

Isometry

Unitary operator

Unitary matrix

Positive definite

Singular values

Norm of a linear map

$\|T\|$

Ball

B

Ellipsoid & Principal axes

$s_1 f_1, \ldots, s_n f_n$

$T(\Omega)$

$P(v_1, \ldots, v_n)$

Parallelepiped & Edge

Box

Volume of a box

Volume

volume Ω

Operators on Complex Vector Spaces

Generalized eigenvector

Nilpotent

Generalized eigenspace

$G(\lambda, T)$

Multiplicity

Characteristic polynomial on $\mathbf{C}$

Block diagonal matrix

Jordan basis

Trace of a matrix

tr

Trace of an operator

Multilinear Algebra and Determinants

Bilinear Form

$V^{(2)}$

Matrix of a bilinear form

$\mathcal{M}(\beta)$

Symmetric bilinear form

$V^{(2)}_{\text{sym}}$

Symmetric matrix

Alternating bilinear form

$V^{(2)}_{\text{alt}}$

Quadratic form

q_β

m-linear form

$V^{(m)}$

Multilinear form

Alternating multilinear forms

$V^{(m)}_{\text{alt}}$

Permutation

$\operatorname{perm} m$

Sign of a permutation

α_T

Determinant of an operator

$\det T$

Determinant of a matrix

$\det A$

Characteristic polynomial

Bilinear functional

$\mathcal{B}(V, W)$

Tensor product

$V \otimes W$

$v \otimes w$

Bilinear map

Inner product on tensor product

m-linear functional

$B(V_1, \ldots, V_m)$

Generalized tensor product

$V_1 \otimes \cdots \otimes V_m$

$v_1 \otimes \cdots \otimes v_m$

m-linear map

Inner product on generalized tensor product